



ROSS CASEBOOK 2012

Ross Consulting Club

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Table of Contents

☐ Introduction & Acknowledgements	2
☐ Firm Overview	4
☐ Industry Overview	18
☐ Case Interview Basics	26
☐ Cases	44

Note to the reader

We are proud present the RCC 2012 casebook. This document is meant to provide a brief overview of case interviews and a series of practice cases. For each case, we have specified the source, difficulty level, and industry. In this case book, the difficulty level for all cases are medium or hard. Therefore, we recommend that you refer to previous Ross casebooks for easier cases if you just started preparing for case interviews.

For the first time, we have added firm and industry overview sections to the RCC casebook. The material is a starting point in your effort to learn more about firms and different industries that could give you some insights while saving you a lot of time. However, it is only a first step and we encourage you to build upon the information by doing your own research on industries and engaging with firms to gain a deeper understanding.

Best of luck in the upcoming recruiting season!

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Special thanks to Bain & Co and Booz & Co for providing full cases which will help Ross candidates gain a better understanding of the firm's expectation in a case interview.



FIRMS OVERVIEW

Accenture

Accenture is a large global management consulting firm specializing in executable solutions and organizational transformation.

About

- Focused on practical solutions that clients can implement
- Balance of strategy and implementation
- Rapidly growing management consulting division
- Internal Strategy College offers consultants the opportunity for professional growth
- New consultants hire into the global operating model or a specific industry

Interview process overview

First Round:

- Two 45- minute interviews consisting of a fit portion and a 30 minute case
- Second Round:
- Three 45- minute interviews consisting of a fit portion and a 30 minute case

Quick Facts

- Number of consultants: 17,000
- Number of offices: 200
- Services: 18

Career path

- Consultant
- Manager
- Senior manager
- Senior Executive

A.T. Kearney

A.T. Kearney is a global team of forward-thinking, collaborative partners that delivers immediate, meaningful results and a long-term transformational advantage to clients and colleagues.

About

- Traditional strengths are strategy and operational consulting
- Increased presence in private equity work
- Diverse work environment and friendly colleagues
- Strongest industry verticals: Industrial and Consumer Retail, Energy, and Public Sector
- Recent Managing Director change and aggressive growth strategy has been announced for the firm

Interview process overview

- First Round
 - Two back-to-back 45 min interviews, each includes fit and case
 - Conducted by a Manager/Principal
- Second Round
 - 15 min email writing test
 - Written case (60 min prep, 30 min presentation)
 - 45 minute fit interview
 - 45 minute full case

Quick Facts

- Number of consultants: 2200
- 57 offices in 39 countries
- 13 Industry groups
- 10 practice areas

Career path

- Associate
- Manager
- Principal
- Partner

Bain & Company

Bain & Company is a global management consulting firm differentiating itself in solving business problems for clients by working with the clients' team as business partners and focusing on results.

About

- Expertise across all major industries and across functions
- Bain redefined the boundaries of traditional strategy consulting in working with companies such as: Tied Economics, BridgeSpan Group, PE consulting, Bain Capital
- Emphasis on people – opportunities to balance work –life, international transfers, externships, private equity rotations
- Leading consulting firm used by major private equity firms

Interview process overview

First Round:

- Two-45 minute interviews including a brief fit interview (5-10min) and a case interview

Second Round:

- 45 min behavioral interview
- 45 minute interviews case interview
- Written case: Interviewee has 60 min to review a written case following 30 min presentation/Q&A by interviewer

Source: www.bain.com

Quick Facts

- Number of consultants: 5500
- Number of offices: 48

Career Path

- Consultant
- Case Team Leader
- Manager
- Principal
- Partner

Booz & Company

Booz & Company is a global management consulting firm known for its functional expertise, industry foresight, and “sleeves rolled up” approach to working with clients.

About

- Booz & Company has always been known for deep industry and functional expertise across public and private sectors, influential global studies and books, and management magazine “strategy + business”
- Emphasis on mentoring and assessment – senior mentor, junior mentor, 360 degree performance assessment
- Expertise across diverse industries and functional areas

Quick Facts

- Number of consultants: 3000
- Number of offices: 58

Career Path

- Associate
- Senior Associate
- Principal
- Vice President, Partner

Interview process overview

First Round:

- Two 45- minute interviews consisting of a fit portion and a 30 minute case

Second Round:

- Two 45 minute interviews with partners following the same format as the first round.

Boston Consulting Group

BCG is a global management consulting firm and one of the world's leading advisor on business strategy. Commitment to both clients' success and its own standards is what sets BCG apart.

About

- Regional Staffing model
- Creative and supportive environment
- BCG provides one of the lowest leverage ratios in the consulting industry; senior management works closely with junior consultants
- Emphasis on a generalist approach. Consultants are not required to specialize in an industry or service line until reaching the Principal level

Quick Facts

- Number of consultants: 5600
- 77 offices in 42 countries
- 19 Industry groups
- 18 practice areas

Career path

- Consultant
- Project Leader
- Principal
- Partner

Interview process overview

- First Round
 - Two Case Interviews (Includes a section for behavioral)
 - Conducted typically by 2nd year Consultants to Principals
- Second Round
 - Three Case Interviews typically conducted by senior representatives
- All Interviews are 45 min with at least 25 min dedicated to the case
- Some offices (such as Chicago) piloting a written case

Deloitte

Deloitte is a global management consulting firm specializing in strategy and implementation across a broad range of industries and functions.

About

- National staffing model provides flexibility to work in a variety of industries regardless of office location
- Focused on executable strategy that offers opportunities in both strategy and implementation
- Job flexibility and culture encourage work-life balance
- Recently invested \$300M in Deloitte University, a leadership center near Dallas
- Pending acquisition of Monitor Group will expand strategy practice and capabilities

Quick Facts

- Number of consultants: 20,000
- Number of offices: 89 US Offices
- Practices
 - Strategy & Operations
 - Technology
 - Human Capital

Interview process overview

First Round:

- 30 min behavioral interview
- 30 min case interview

Second Round:

- 30 min behavioral interview
- 60 min individual case interview
- 90 min group case interview

Career path

- Senior Consultant
- Manager
- Senior Manager
- Partner or director

Ernst & Young

Ernst & Young (EY) is one of the largest professional service firms in the world service delivering capabilities to help companies turn innovation into action, information into insight and risk into results.

About

- One of the Big Four Accounting +consulting firms
- Prominent Advisory practices are
 - PI (Performance Improvements, includes all operations related work and a strategy function known as “strategic direction
 - TAS (Transaction Advisory Services) – includes Commercial Advisory (valuation, due diligence) and Operational Transformation Services (Post Merger Integration)

Quick Facts

- Number of consultants: 167,000 ¹

Career Path

- Senior Consultant
- Manager
- Senior Manager
- Partner

Interview process overview

First Round:

- On campus, consisting of two sessions, a behavioral interview and a case interview

Second Round:

- Behavioral interview
- Case interview

McKinsey & Company

McKinsey & Co is a large global management consulting firm focusing on high profile studies for businesses, governments, and institutions.

About

- Strong research department supporting consultants
- National staffing model
- High profile clients and studies
- Teams with diverse backgrounds (MBAs, PhDs, JDs)
- A culture that promotes work-life balance
- Encourages active discussion; individuals have “obligation to dissent”

Interview process overview

Selected candidates have to take Problem-Solving test: 26 questions, 60 min

First Round:

- Two 45-60 minute interviews consisting of a case and fit questions

Second Round:

- Three 45-60 minute interviews consisting of a case and fit questions

Note: Fit questions address three main points: personal impact, leadership, and problem solving skills

Quick Facts

- Number of consultants: 17,500
- Number of global offices: 99
- Industry Practices: 22
- Functional Practices: 8

Career path

- Associate
- Engagement Manager
- Associate Principal
- Partner
- Director

Oliver Wyman

Oliver Wyman is one of the world's fastest-growing strategy consulting firms with specialized expertise in strategy, operations, risk management, organizational transformation, and leadership development.

About

- Specialized industry and functional expertise
- Rigorous, proven strategy and operational improvement methodologies
- Collaborative working style
- Agenda-setting research
- Work-life balance

Interview process overview

First Round:

- One behavioral interview – 45 min
- One case interview – 45 min

Second Round:

- Behavioral interview - 1 hr
- Case interview – 1 hr

Quick Facts

- Number of consultants: 2000
- Number of offices: 20
- Career Tracks
 - Financial Services consulting
 - General Management consulting
 - Organization transformation

Career Path

- Associate
- Senior associate
- Job Manager
- Senior Job Manager
- Associate Partner
- Partner

PwC Advisory

PwC Advisory is a rapidly growing consulting organization backed by the stability and strength of the PwC brand. They support clients in designing, managing and executing lasting beneficial change.

About

- National Staffing model
- Focus on four industry verticals: Financial Services, Health Care, Product and Services, Public Sector
- Consulting practice projected to double in the next two years
- High investment by the firm on internal networking events to develop strong intra-company bonds
- New employees recruit for a specific industry focus rather than a generalist role

Interview process over view

- First Round (On-campus, 2 sections, 45 min each)
 - Behavioral
 - Mini-Case and Fit
- Second Round (3 sections, 45 min each)
 - Behavioral
 - Content interview: Industry Focus Interview
 - Case interview: a presentation based on a case emailed 48 hours in advance

Quick Facts

- Practice areas:
 - Strategy
 - Finance
 - Operation
 - People & Change
 - Risk

Career path

- Senior Associate
- Manager
- Principal
- Partner

Roland Berger

Roland Berger is one of the top international consultancies. It prides itself on developing creative strategies and implementing practical solutions.

About

- Values entrepreneurial spirit and individuality of consultants
- Three core values: entrepreneurship, partnership and excellence
- Deep Understanding of diverse cultures and markets
- Internal transfer policy that allows consultants to permanently change offices if they have suitable language skills
- Every new hire is assigned a senior mentor to help with the transition and personal development

Interview process overview

- First Round
 - 45 min interview which includes fit and short case
 - Conducted by 1-2 consultants
- Second Round
 - Personal fit interview
 - Business knowledge interview
 - Case interview that includes a presentation
 - Conducted by consultants and members of management team

Quick Facts

- Number of consultants: 2700
- 49 offices in 35 countries
- 25 Industry groups and 26 functional areas worldwide
- 3 industry groups and 4 functional areas in North America

Career path

- Senior Consultant
- Project Manager
- Principal
- Partner

The Cambridge Group

The Cambridge Group is a management consulting firm specializing in helping clients identify and capture market demand.

About

- Acquired by The Nielsen Company in 2009 enables access to key customer insights
- Emphasis on helping Growth strategy – top line growth
- Major industries include retail, CPG, and financial services
- Office-based consulting model limits frequency of travel

Interview process overview

Single Round:

- 3 consecutive interviews totaling 2 hours consisting of a case and fit questions.

Quick Facts

- Number of consultants: 115
- Number of offices: 1 (Chicago)

Career path

- Consultant
- Project Manager
- Project Director
- Principal

ZS Associates

ZS Associates is a global management consulting firm specializing in using data driven strategies to provide sales and marketing solutions.

About

- Expertise in marketing and sales with a focus in healthcare
- Partnership with clients to design and implement solutions
- ZS services include consulting, outsourcing, technology, and software
- Project-specific and formal training provide opportunities for continued professional development

Quick Facts

- Number of consultants: 2000
- Number of offices: 20
- Practices
 - Business consulting
 - Business operations
 - Business technology

Career path

- Consultant
- Manager
- Associate Principal
- Principal

Interview process overview

First Round:

- On campus consisting of two sessions, a behavioral interview and a case interview
- Second Round:
 - Behavioral interview
 - Case interview
 - Business presentation interview – evaluation of communication skills

Source: www.zsassociates.com/about.aspx



INDUSTRY OVERVIEW

Airline Industry

Revenue Streams

- Ticket sales
- Charges for bags and on-board services
- Fuel surcharges
- Capacity optimization is key for profitability

Cost Structure

- Labor – 11%
- Fuel – 32%
- Aircraft depreciation/rentals - 12%
- Airport gates/ facility rentals – 6%
- Maintenance – 5%
- Weather resulted in cancelled flights can be a significant expense

Market Trends

- Airlines are consolidating
- Proliferation of low cost carriers offering pay per service options
- Flexible capacity is an important factor in airline profitability
- Average annual growth: 5.1%
- Online booking and checking to reduce administrative expenses
- Profit margins around 1.6-2.5%

Customer Segments

- Leisure travelers – 71%, generally price sensitive
- Business travelers – 25%
- Freight Transportation – 4%

Channels

- Online travel agencies (Orbitz, Expedia, Kayak, etc.)
- Airlines' sales team: call centers, online, or kiosk
- Travel management companies (TMCs) serving corporate clients

Risk

- Changes in fuel prices have a major impact on profitability
- Economic conditions greatly impact amount of leisure travelers

Misc.

- Highly susceptible to threats of terrorist activity
- Disease outbreaks also limit airline demand

Auto Industry

Revenue Streams	Cost Structure	Market Trends
▪ Vehicle sales ▪ Vehicle financing ▪ Extended Warranties ▪ Branding is key for selling similar product at differing price points	▪ Materials – 78%, Long term contracts with key suppliers are common ▪ Labor – 5%, Relationship with the UAW is critical ▪ Marketing – 3.6% ▪ Depreciation – 2.1%	▪ Focus on fuel improving efficiency ▪ Increased popularity of hybrid and electric vehicles ▪ Top 8 firms control 80% of the market ▪ Average annual growth: 3.3% ▪ Profit margin around 2.4% ▪ Significant model redesigns occur approximately every 5 years
Customer Segments	▪ Large and midsize ▪ Small and compact ▪ SUVs and light trucks	
Channels	▪ Dealership networks – largest domestic segment ▪ Wholesalers – purchase fleets of vehicles for businesses and resale to dealerships ▪ Rental companies	
Risk	▪ Relationship with United Auto Workers (UAW) union a major factor in industry profitability	
Misc.	▪ Purchase decisions influenced by current fuel prices ▪ Plant capacity utilization is very important, over 80% utilization required for profitability	

Commercial Banking Industry

Revenue Streams

- Loan interest
- Fee based services
- Credit cards
- Revenue is dependent on prime rate and aggregate household debt

Cost Structure

- Wages – 26%
- Bad debt expense – 21%, recent levels are significantly higher than historical average (5%)
- Interest expense – 15%
- Facilities – 7%

Market Trends

- Industry consolidation, especially in the wake of subprime mortgage crisis
- Increased government regulation
- Mobile banking is increasing more important to long term success
- Average annual growth: 5.2%
- Profit margin: 20%

Customer Segments

- Retail customers – 56%
- Corporate clients – 39%

Channels

- Branch offices
- Stand alone ATMs
- Mobile banking

Risk

- Non traditional competitors are beginning to offer banking services - Retail Co.

Misc.

- New legislation limits banking fees and requires banks to maintain higher capital reserves, both have decreased industry profitability

Pharmaceuticals

Revenue Streams

- Pharmaceutical manufacturing industry is highly capital intensive with high revenue volatility
- The effect of seasonality is high on certain products (vaccines and cold medicine) and low on other products (pain medicines)
- Federal government grants for R&D

Cost Structure

- The manufacturing cost of pharmaceuticals is the largest share of the industry's cost - 31.5%
- Research & Development- 20%
- Marketing costs – 7%
- Wages account for 11% in the cost structure
- The industry also faces high liability insurance costs and high legal fees and settlements

Market Trends

- Healthcare reform is expected to boost sales as more individuals gain prescription drug coverage in 2014
- Tariff barriers are no longer a relevant form of protection
- Demographic shifts will increase sales over next five years to 2017
- Loss of patent on key drugs for many large pharma companies. around 2010-11

Customer Segments

- Patients/consumers who need these drugs/medicines
 - The targeting strategy has to consider two other stakeholders: 1. Doctors who prescribe these medicines 2. Insurance companies that pay for them

Channels

- Over-the-counter (can be sold without prescription): Retail outlets – CVS, Walgreens; Mail order
- Prescription drugs: Hospitals, pharmacies
- Mail order pharmacy: Express Scripts, Walgreens

Risk

- Price competition from generic drug manufacturers
- Change in drug utilization and mix patterns
- Government healthcare policies that are not favorable

Misc.

- Key success factor : Blockbuster Drug (>\$1B in annual sales)
- FDA approval required before any sales can be made

Retail

Revenue Streams

- Women's Apparel sale accounts for the major share of revenues at 21%
- Drugs & Cosmetics and Furniture & Household appliances contribute 16% each to total revenue
- Children and Men's wear are each 10% of retail revenue
- Toys - 8%, Footwear – 5% and misc. items - 13% are other sources of revenue

Cost Structure

- Industry costs are similar for most operators, but vary between firms of different operating sizes and specialties
- Largest expense for the industry is Cost of Goods Sold which has increased due to weak sales
- Employee wages account for 13%
- Rental costs are 5%
- Marketing costs are close to 10%

Market Trends

- Over the five years to 2017, retail sector revenue is forecast to grow at an average rate of 4.5% per year
- Employment in the industry is projected to grow slowly
- Consolidation is expected to continue within this industry
- The industry's contribution to the GDP is expected to underperform compared to the US economy

Customer Segments

- The industry is consumer-oriented and, due to the spectrum of products, its markets are generally segmented into different age groups
- Segment shares of the revenue have remained relatively unchanged over the past five years

Channels

- Big-Box Retailers
- Specialty Retail stores
- Discount Department stores

Risk

- Changes in consumer disposable income can cause consumers to defer purchasing products from retailers as industry is sensitive to changes in economic activity

Misc.

- Major players are Retail Co. , Sears, Macy's, Target and JC Penny. Together these companies account for 81% of market share

Telecommunication

Revenue Streams

- Plain old telephone calls, and increasing text and images. High-speed internet access, broadband information services and interactive entertainment
- The fastest growth comes from services delivered over mobile networks.
- Advertising income accounts for about 5% of total industry revenue.

Cost Structure

- Investment and investment-related costs are 65-70 percent of the costs of production
- Wage shares are at about 25%
- A notable part of the investments are what economists refer to as “sunk costs”. These are long term investments which can be used only for specific economic activities.

Market Trends

- Expectations of always-on service everywhere forcing operators to boost network capacity and connectivity
- Increased demand of a variety of new services like mobile payment platforms and cloud computing
- Revenue increase forecast for Internet services is 7.9% per annum till 2017 and for wireless telecom to increase by 4.8% till 2017

Customer Segments

- Residential and small business markets
- The corporate market - less price-sensitive than residential customers. Large multinationals, premium service buyers include those opting for high-security private networks and videoconferencing.

Channels

- Retail Channels
- Direct Sales

Risk

- Capital-intensive telecom industry the biggest barrier to entry is access to finance
- New technology is prompting a raft of substitute services, with lower prices and more exciting services driving down industry profitability

Misc.

- Average Revenue Per User (ARPU): Measure of growth performance.

Utilities

Revenue Streams

- Transmitted electricity is separated into two categories, base load and intermittent electricity
- Base load electricity is expected to account for the bulk of industry revenue [95%] (Coal – 36%, Natural gas – 25%, Nuclear – 17%, Others – 17%)
- Intermittent electricity is generated from renewable energy sources [5% revenue share]

Cost Structure

- Purchased power accounts for the largest component of this industry's cost structure – 44%
- Wages - 9% , wage growth has fallen due to slower rate increases and industry consolidation
- High depreciation costs because infrastructure requires significant capital investments
- Marketing, maintenance contracts and other costs – 15%

Market Trends

- Overall growth over the next five years is anticipated to be positive
- In the short term, the recession's lingering negative effects are anticipated to constrain growth
- Renewable power, such as solar and wind, are projected to grow strongly over the next five years
- Public utility commissions (PUC) are expected to grant rate increases, fueling revenue and profit growth.

Customer Segments

- Residential
- Commercial
- Industrial and Transportation

Channels

- Transmission grids

Risk

- The Utilities industry is in the mature stage of its life cycle and is associated with low risk s
- Seasonality of electricity consumption (due to weather shifts) can squeeze revenues in the short term

Misc.

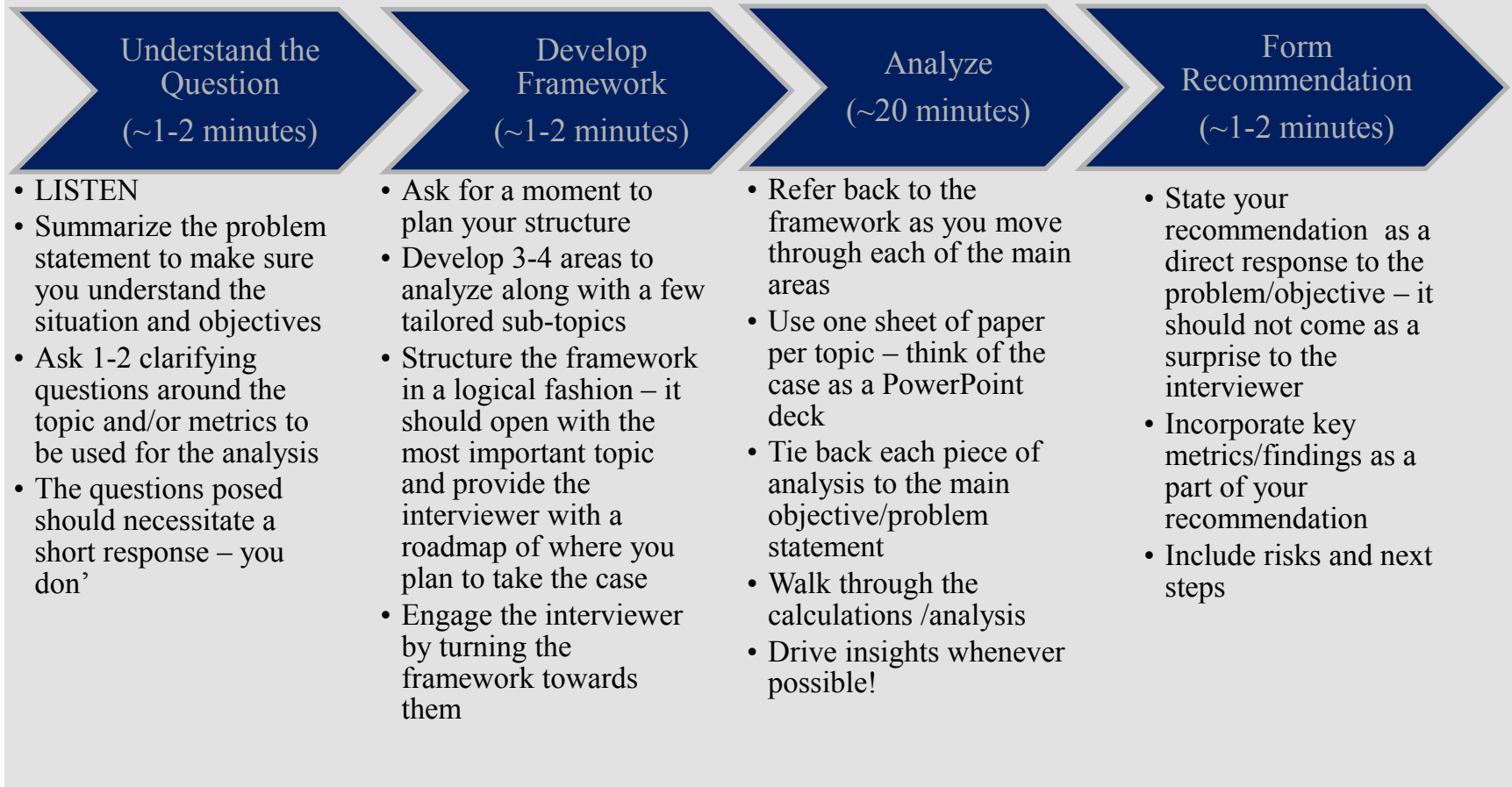
- Advanced metering technology will become a consistent feature of this industry over the next five years



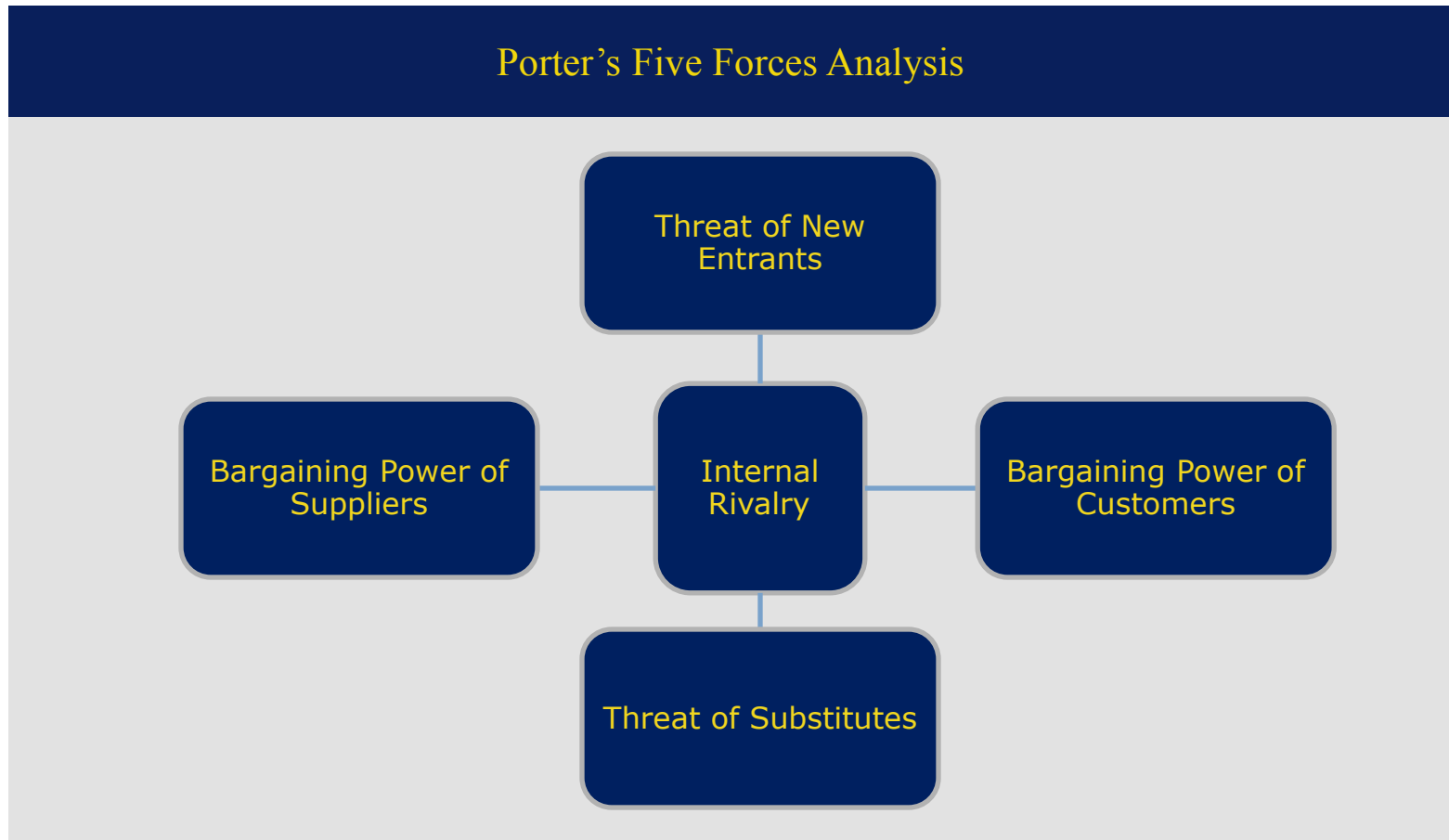
Case Interview Basics

Case Structure

The overall structure of the case interview takes the following form:



Porter's Five Forces



Porter's Five Forces

Concept	Key Drivers
Internal Rivalry	- Concentration and balance - Industry growth - Product differences - Exit barriers - Overcapacity
Threat of New Entry (Barriers to Entry)	- Economies of scale - Capital requirements - Access to distribution channels - Competitor response - Brand identity - Proprietary product differences
Threat of Substitutes	- Switching costs - Relative pricing - Availability of and consumer propensity to substitute products
Bargaining Power of Suppliers	- Supplier concentration - Switching costs - Threat of forward integration - Product differentiation
Bargaining Power of Customers	- Buyer concentration - Buyer volume - Buyer switching costs - Ability to backward integrate - Substitute products

Key Marketing Concepts

4Ps	Considerations
Product	Features and capabilities Quality and reputation Service and warranties Packaging and size Positioning and market segmentation Differentiated versus commodity
Promotion	“Pull” versus “push” Consumer awareness Loyalty Advertising medium Public relations Buying process Trial/Repurchase
Price	Perceived value Willingness to pay Retail/Discounts Economic incentives Skimming Strategy → relation to market size, product lifecycle, and competition
Place (Distribution)	Channels Coverage Inventory → levels, turnover, carrying costs Transportation → alternatives, efficiencies, costs

Key Marketing Concepts

3Cs	Considerations
Company	Strengths/Weaknesses/Opportunities/Threats Strategy and vision Available resources/Capacity Experience/Learning Curve Financial Culture/Organizational structure
Competition	Industry Size/Number/Market share Economies of Scale/Scope Capabilities/Experience Resources → financial, distribution
Customer	Perceptions Loyalty Switching costs Purchase behavior Segmentation Market characteristics/trends

To make this a 5Cs analysis, one would also evaluate costs and channels. Data for these two dimensions is covered elsewhere in the casebook.

General Frameworks

Topic	Key Drivers
Revenue	Volume • Internal → Price, Customer Service, Distribution/Inventory/Capacity • External → Competition, Substitutes/Complements, Market Forces/Demand Price → Competition, Elasticity, Differentiation, Segments Product Mix → Attributes (e.g. niche, patent), Quality, % of Revenue, Variety Alternative Revenue Streams Number of Stores
Costs	Fixed Costs → Manufacturing, Labor, Marketing, Overhead, IT, SG&A, PP&E Variable Costs → Inputs, Distribution, Marketing, Maintenance, Packaging, Inventory Balance Sheet Items Benchmark Opportunity Cost/Cost Accounting/Capacity Utilization External → Union strikes, Technology, Currency Fluctuations, Tariffs, De-/Regulation
Competition	Rivals (structure) New Entrants Substitutes Reaction Position

General Frameworks

Topic	Key Drivers
Customers	Market Size Segments Needs Purchase Drivers Price Elasticity Retention/Loyalty
Processes	Manufacturing Marketing Sales Distribution Customer Service IT R&D Forecasting
Company	Core Competencies Cost of Capital Brand Organization / Incentives Controls Financial Capability Management Capability

General Frameworks

Topic	Key Drivers
Macro	Legislation Unions Technology Economy → Oil, Interest Rates, Unemployment International Issues → Politics, Regulations, Taxes, Tariffs Environment Socio-Cultural Demographics
Supply Chain	Suppliers Distributions
Industry	Barriers to Entry/Exit Lifecycle Consolidation Government Policy Capital Costs Access to Technology, Distribution, etc.

Key Formula Review

Topic	Formula
Time Value of Money	$\text{Value to Perpetuity} = \frac{\text{Value of Asset}}{\text{Discount Rate}}$ $\text{NPV} = \sum_{t=0}^n \frac{\text{Annual Cash Flow}}{(1+r)^t}$
Rule of 72	$\text{Time for Invested Principle} = \frac{72}{r}$ • r = rate of return • At 7% r, the investment will double every 10 years • At 10% r, the investment will double every 7 years
Little's Law	$\text{Inventory} = \text{Throughput} \times \text{Flow Time}$
Inventory	$\text{Inventory Turns} = \frac{\text{COGS}}{\text{Average Inventory}}$ $\text{Days of Inventory} = \text{Inventory Turns} * 365$
Profitability	$\pi = Q(P - VC) - FC$
Breakeven	$\text{Breakeven} = \frac{FC}{P - VC}$
Margin	$\text{Gross Margin} = \frac{P - C}{P}$ $\text{Net Margin} = \frac{\text{Net Income}}{\text{Sales Revenue}}$
Markup	$\text{Markup} = \frac{P - C}{C}$

Key Formula Review

Topic	Formula
Return on Assets (ROA)	$\text{Return on Assets} = \frac{\text{Net Income}}{\text{Total Assets}}$
Return on Equity (ROE)	$\text{ROE} = \frac{\text{Net Income}}{\text{Total Shareholders' Equity}}$
DuPont Analysis	$\text{ROE} = \frac{\text{Net Profit}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Assets}} \times \frac{\text{Assets}}{\text{Equity}}$ <i>ROE = operating efficiency × asset efficiency × financial leverage</i>
Working Capital	$\text{ROI} = \frac{\pi}{K}$ • K = Capital Invested (e.g. Assets, Working Capital) • Working Capital = Assets – Liability
Income Statement	Sales – COGS = Gross Profit – SG&A = EBITDA – Depreciation/Amortization = Operating Profit – Interest Expense = EBIT – Tax Expense = Net Income

Economics Review

Concept	Definition
Adverse Selection	Situation in which an individual's demand for insurance is aligned to their risk of loss (i.e. people with the highest expected value will buy insurance) and the insurer cannot account for this correlation in the price. • Restrict choice • Equalize information • Signaling
Consumer Surplus	Economic gain achieved when consumers purchase a product for a price less than their willingness to pay. • $\text{Consumer Surplus} = \text{Willingness to Pay} - \text{Price}$
Economies of Scale	The average cost per unit for a business entity is reduced by increasing the scale of production.
Economies of Scope	The average cost for a business entity is reduced by producing two or more products.
Elasticity	• If $E > 1$, decrease price to increase revenue • If $E < 1$, decreased price leads to lower revenue
Insurance	Form of risk management used to hedge against the risk of a loss in which the cost is equal to expected loss.
Law of Diminishing Returns	At some point in the production process, the addition of one more unit of output, while holding everything else constant, will eventually lead to a decrease in per unit returns.
Marginal Cost	Cost of one more unit of output.

Economics Review

Concept	Definition
Monopoly	Entity is the only supplier of a particular good. - Lack of competition → produce less and charge more - Barriers may include government regulation, networks, patents, etc. - Revenue is the midpoint of the demand curve
Moral Hazard	The unobservable actions and risks that humans may take once a contract is signed since they don't bear consequences. It is a special case of information asymmetry that affects the cost of transaction.
Oligopoly	Market is dominated by a small number of sellers. - Dominant strategy is always better - Sequential games – commitments help
Perfect Competition	- Firms take price → $MR = P$ - Maximum profit = $MR = MC$ $P < AVC$ → shut down
Price Discrimination	Situation in which identical goods are sold at different prices from the same provider. 1st degree → Different price for different willingness to pay 2nd degree → Different price for different quantities 3rd degree → Different price for different segments (attributes)
Risk Averse	Individuals who prefer certainty over the uncertain for the same expected value (EV).
Risk Neutral	Individuals who are indifferent on risk taking if the EV is the same.
Risk Seeking	Individuals who prefer risk even if the EV for a certain event and the risk is the same.

Glossary

Term	Definition
Arbitrage	The purchase of securities on one market for immediate resale on another market in order to profit from a price discrepancy.
Break-Even	Total amount of revenue needed to offset the sum of a firm's costs. Implies that the firm's profit will be \$0.
CAGR	Compound Annual Growth Rate: $(\text{Ending value}/\text{beginning value})^{(1/\# \text{ of years})}-1$. Most likely to show up in a case with graphs and exhibits.
Capacity	The maximum level of output of goods and/or services that a given system can potentially produce over a set period of time.
Competitive Advantage	When a firm is able to deliver benefits equal to competitors but at a lower cost OR able to deliver greater benefits than competitors.
Contribution Margin	$C=P-V$, where P is unit price, and V is variable cost per unit.
Core Competencies	The activities that a firm does well to create competitive advantage.
Customer Segmentation	Subdivision of a market into discrete groups that share similar characteristics.

Glossary

Term	Definition
Discount Rate	Also known as cost of capital. There is an opportunity cost associated with every investment, with the cost being the expected return on an alternate investment.
Entering New Market	Three main methods: start from scratch, form joint venture, acquire an existing player.
Five Cs	Company, Customer, Cost, Channels, Competition
Fixed Costs	Costs that do not change with an increase or decrease in the amount of goods or services produced.
Four Ps	Product, Price, Promotion and Place
Gross Margin	A Company's total sales minus its cost of goods sold, divided by the total sales revenue, expressed as a percentage.
Horizontal Integration	The acquisition of additional business activities at the same level of the value chain.
International Expansion	Main mechanisms: exporting, licensing, franchising, joint venture, foreign direct investment (acquisition or startup).

Glossary

Term	Definition
Inventory Turnover	A ratio showing how many times a company's inventory is sold and replaced over a period. Should be compared to industry averages: low turnover implies poor sales or excess inventory; high ratio implies either strong sales or ineffective buying.
Learning Curve	Visually shows how new skills or knowledge can be quickly acquired initially, but subsequent learning becomes much slower. A steeper curve indicates faster, easier learning and a flatter curve indicates slower, more difficult learning.
Market Share	The percentage of market size controlled by an individual firm.
Payback Period	The length of time required to recover the cost of an investment.
Market Size	Total size of a population (usually measured in number of people or actual dollar value) that would purchase a company's goods or services. Market size is always relevant and is a question that should be asked.
Product Lifecycle	Four main stages: market introduction, growth, maturity, decline.
NPV	The difference between present value cash inflows and present value cash outflows.
Product Mix	Total number of product lines that a company offers to its customers. Often an important area to explore in profitability cases to identify loss-making products.

Glossary

Term	Definition
Porter's Five Forces	Buyer Power, Supplier Power, Threat of new entrants, Substitutes, Internal Competition. Used for evaluating markets. Also key to think about complements even though that's not mentioned by Porter.
Profit	Revenue minus cost.
Promotion	Coupons, discounts, trials, etc. designed to increase sales of a product or service.
Rule of 72	Also known as the rule of 70, AKA rule of 69. Simply put 72, 70 or 69 in the numerator and the projected annual growth rate in the denominator to give you the amount of time until the investment doubles.
Sales per Square Foot	The average revenue a business creates for every square foot of sales space. Used in the retail industry as a measure of efficiency.
Same Store Sales	A statistic used in retail industry to determine what portion of new sales has come from sales growth and what portion from the opening of new stores.
SWOT Analysis	Strengths, Weaknesses, Opportunities and Threats. Very basic framework, probably not a good idea to put down as your case framework, but good to have as a mental checklist.

Glossary

Term	Definition
Synergies	The idea that the value and performance of two companies combined will be greater than the sum of the separate individual parts. Used mostly in M&A.
Value Chain	Another concept from Michael Porter. His Value chain: Inbound Logistics, Operations, Outbound logistics, Marketing and Sales.
Variable Costs	Costs that vary depending on a company's production volume; they rise as production increases and fall as production decreases.
Vertical Integration	Degree to which a firm owns its backward suppliers or forward buyers.
Weighted Average	An average in which each quantity is assigned a weight. These weightings determine the relative importance of each quantity on the average.



Cases

Case List

1. LawnCo – Bain Original	46
2. Credit Card Processor – Bain Round 1	56
3. Little Bud Co – Bain Final Round	63
4. Digital Expansion Prescott Publishing – Booz Original	73
5. V-Formulae – BCG Round 1	79
6. Retail Co – Accenture Talent and Organization Round 1	88
7. Melsen Medical Scanner – McKinsey Final Round	92
8. Bolsen Pharma – Bain Final Round	97
9. African Call Center – McKinsey Round 1	104
10. Burger Chain – Ross Original	112
11. Fitness Startup Kiosks – Ross Original	118
12. Barley Inc. Cosmetics – Ross Original	124
13. PE Services Firm – Booz Round 1	131
14. PD Gas Buyout	136
15. Ocular Implant – BCG Final Round	143
16. Shipping Company – McKinsey Final Round	146
17. Seaworthy Floating Hotel – Ross Original	155
18. Big Trucks – McKinsey Final Round	165
19. Delson Hard Drive – BCG Final Round	173

Lawn Co. Case Overview

Original case and solution from Bain & Company

Situation

Lawn Co. specializes in residential lawn fertilization, weed control, and disease prevention

Lawn Co. is a large US residential lawn care company with ~20% market share

Lawn Co is highly profitable with margins driven through an aggressive focus on cost

Complication

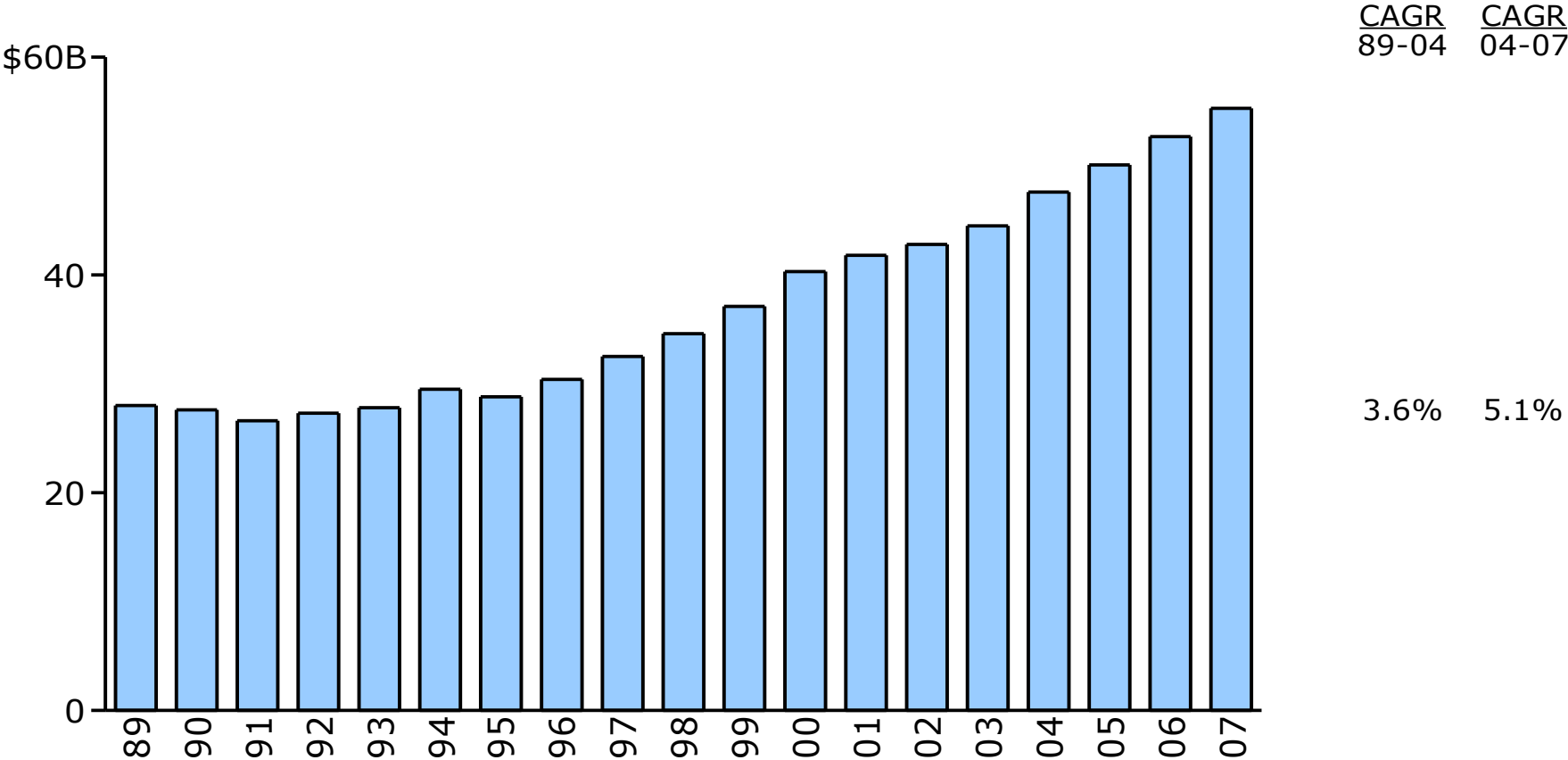
However, Lawn Co. has seen little top line revenue growth in recent years

Key Question

Should Lawn Co. focus on organic growth or should it pursue inorganic acquisitions to grow?

The lawn and landscaping market has been growing at ~5% p.a. since 2004

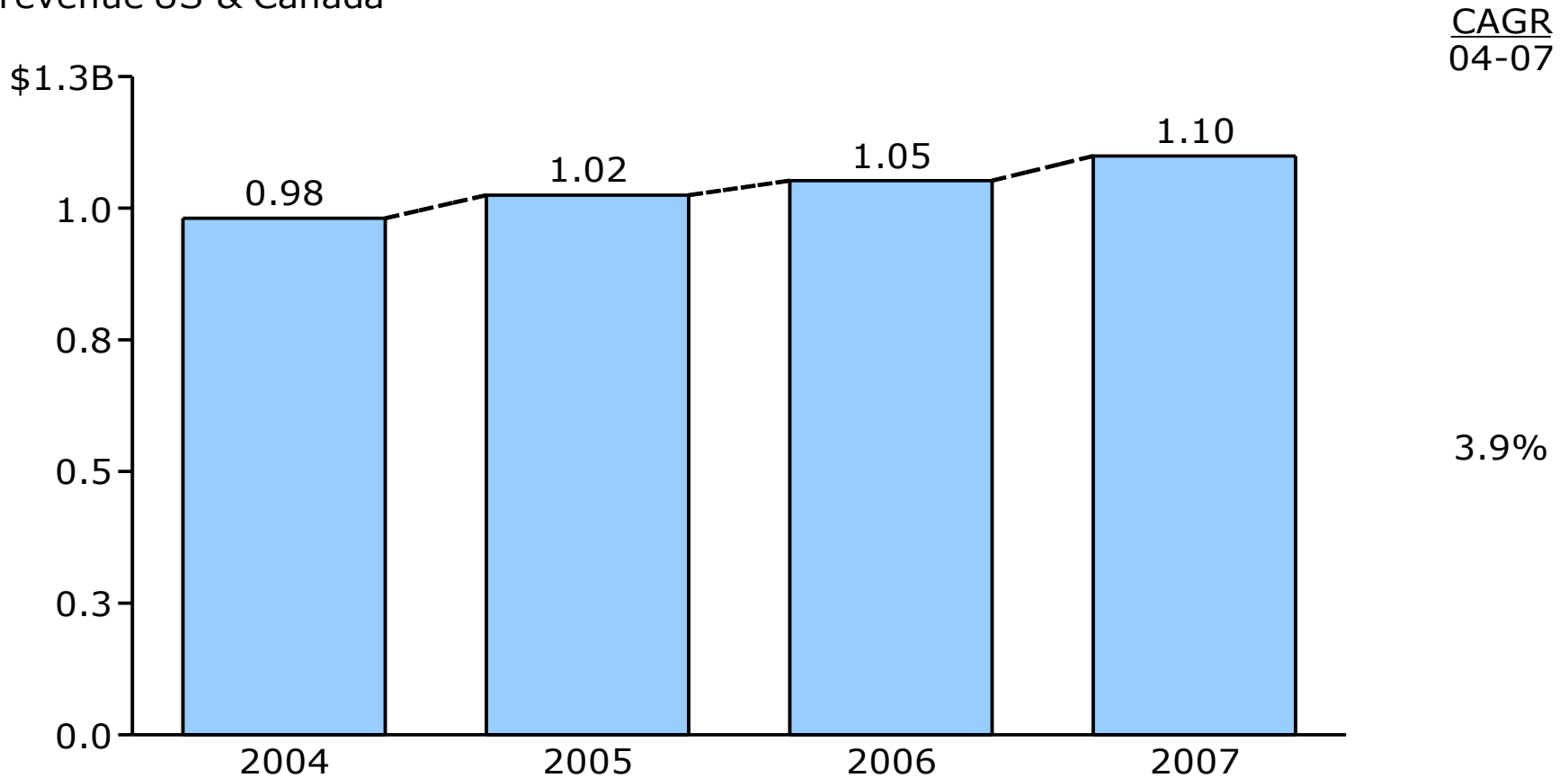
Lawn & Landscaping Market
(Residential & Commercial)



Source: Bain & Co.

Lawn Co. has grown at ~4% p.a. since 2004

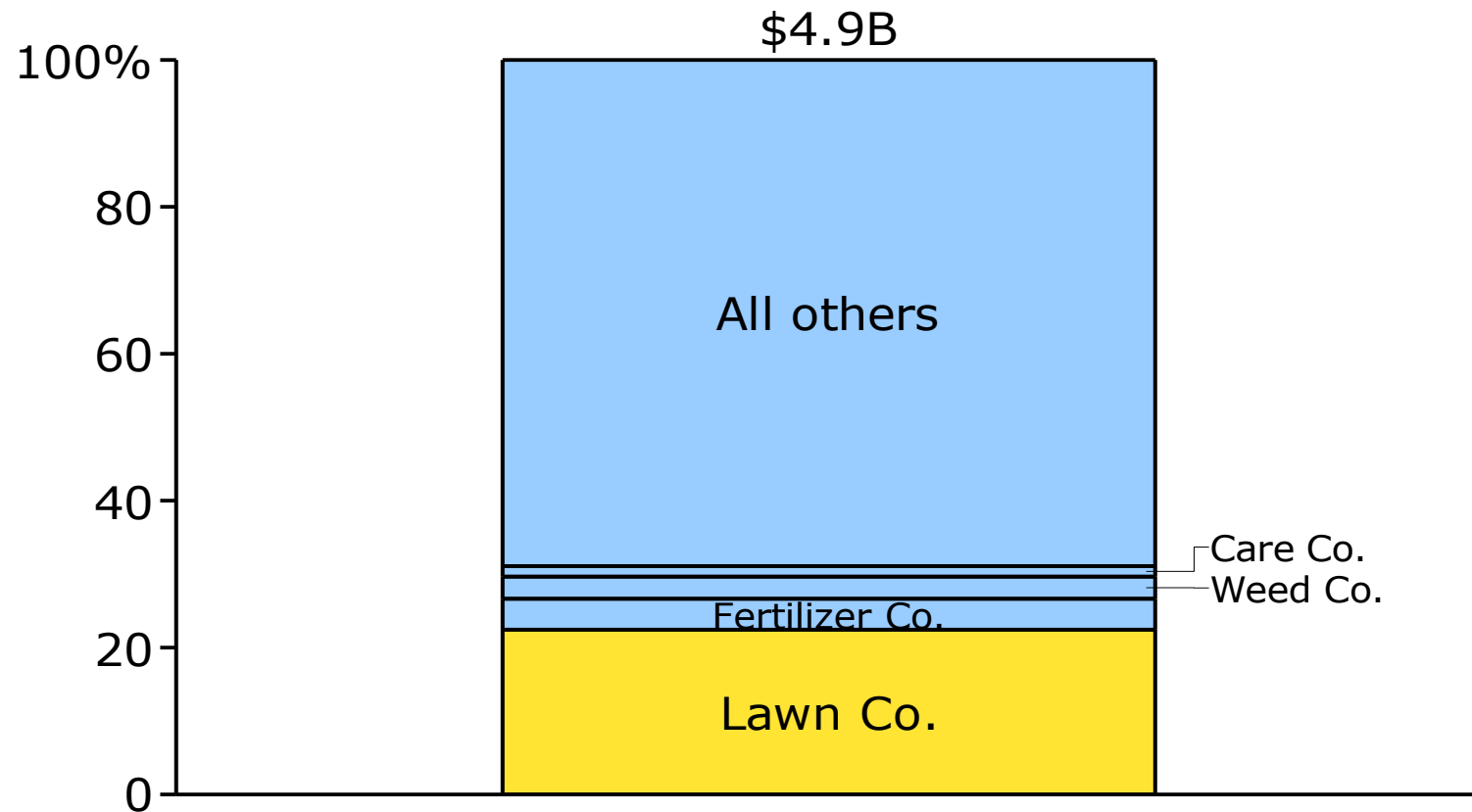
Total revenue US & Canada



Source: Bain & Co.

Lawn Co. leads the lawn services market with ~20% share

Percent of market revenue (2008)

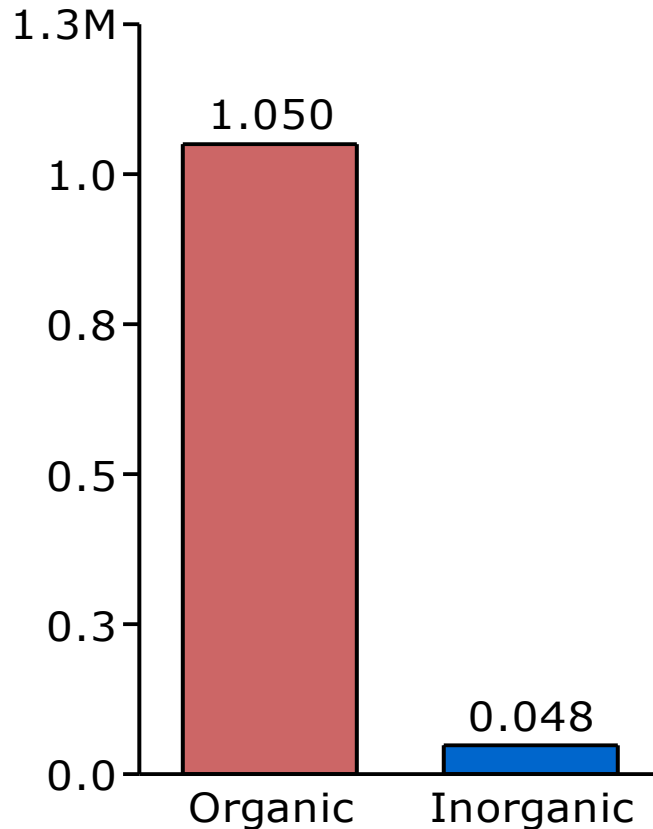


Source: Bain & Co.

In 2007, Lawn Co spent a total of \$120M to acquire ~1.1M new customers

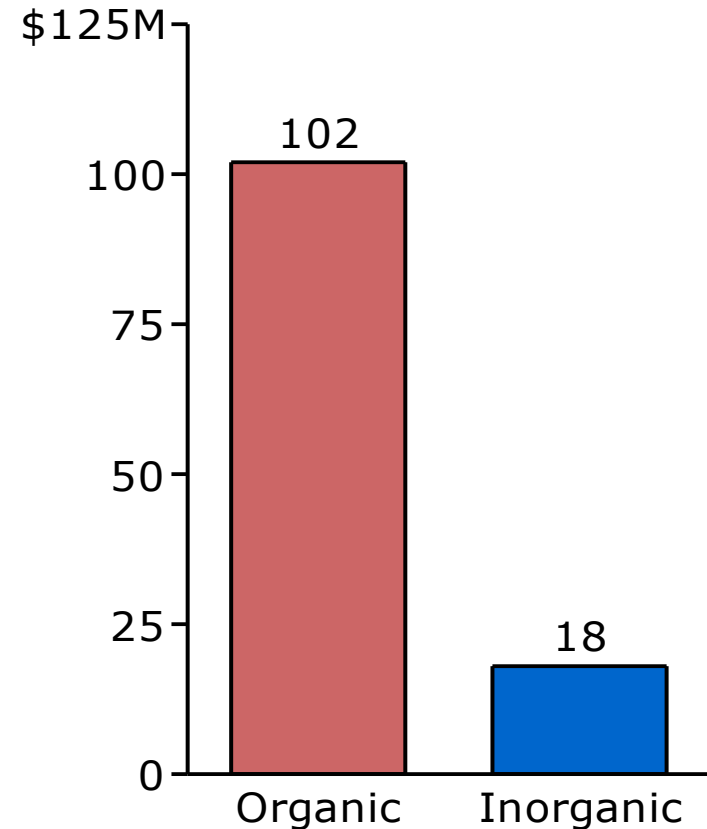
New customers in 2007

Number of new customers



2007 customer acquisition costs

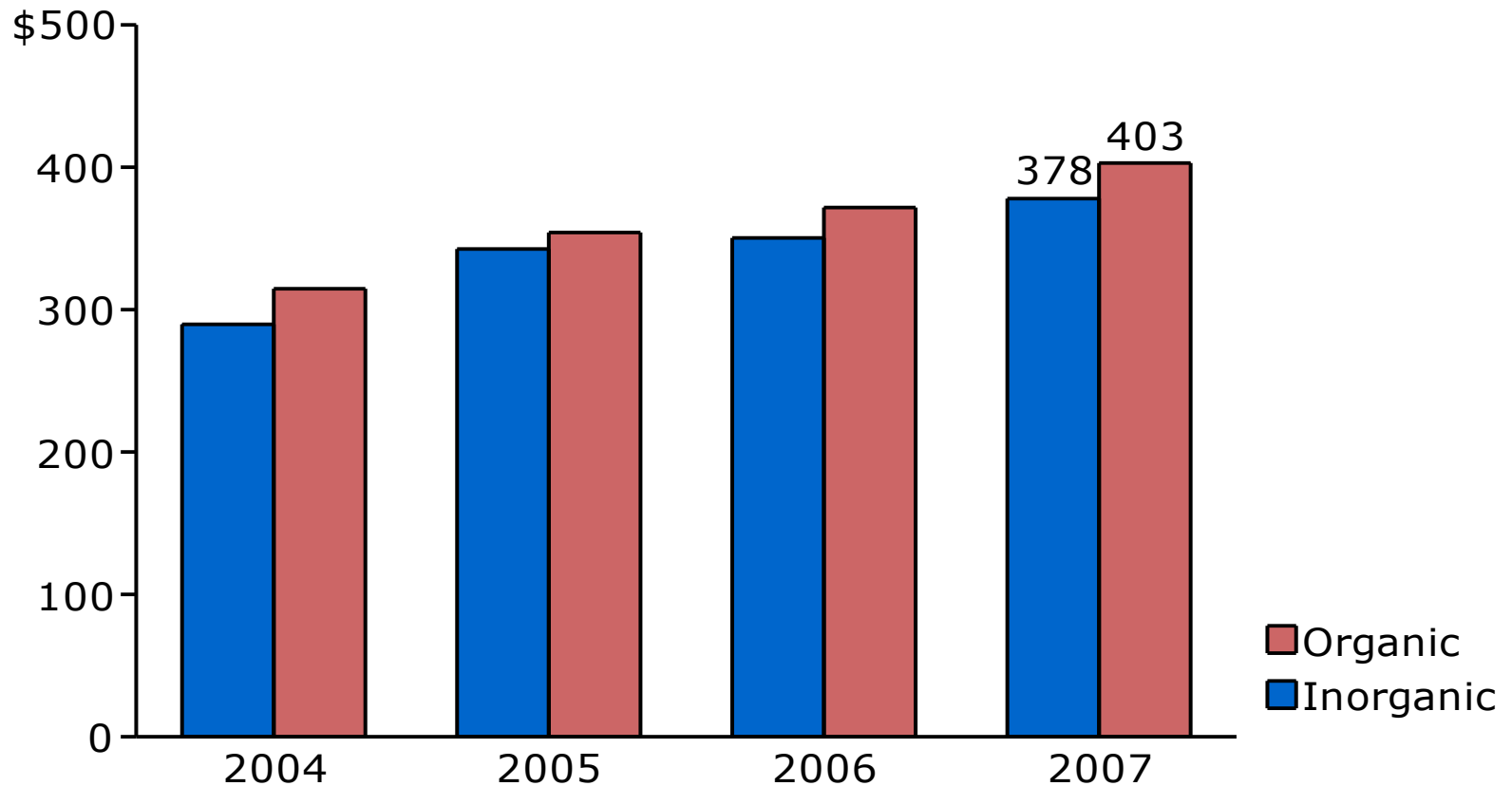
Total cost to acquire new customers



Source: Bain & Co.

Customers acquired inorganically spend less per year than organic customers

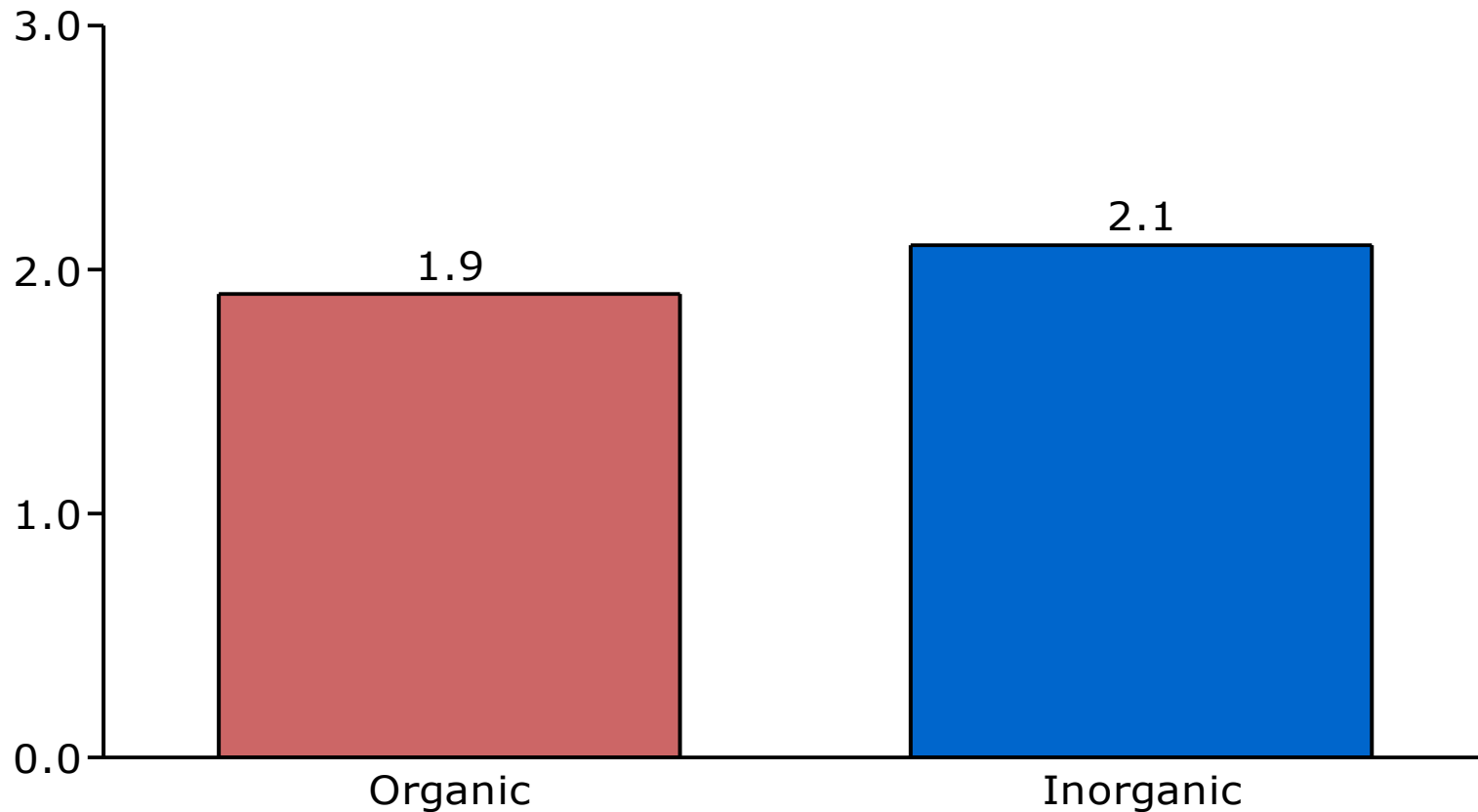
Average spend per year



Source: Bain & Co.

Customers acquired inorganically stay Lawn Co. customers for slightly longer

Average customer lifetime (years)



Source: Bain & Co.

Interviewer Guide: Case Overview

High-Level Model Answer

1. Understanding customer lifetime value is key: Though the interviewee does not need to know the name of the concept, it is crucial he/she understands that each option's attractiveness depends on the cost to acquire and the contribution they will bring in
2. Calculating the cost to acquire each type of customer is the first step. Interviewees will see this is driven by the purchase price per customer acquired for acquisitions and total marketing spend divided by new customers for organic customers. A reasonable answer is \$360 for acquired customers and \$100 for organic customers.
3. The interviewee should then examine the revenue side of each customer. The interviewee should ask if both types of customers spend the same amount and stay for the same length of time. A reasonable answer is \$760 for acquired customers and \$800 for organic customers.
4. Finally, the interviewee should assess the net benefit of each type of customer. Subtracting the acquisition cost from the potential revenue yields that organic customers are more attractive. A reasonable answer is \$700 for organic customers and \$400 for inorganic customers

Other

Key Concepts Tested

1. Profit tree
2. Customer lifetime value

Required Calculations

1. Cost to acquire for organic vs. inorganic
2. Spend for organic vs. inorganic
3. Profit or contribution for organic vs. inorganic

Interviewer Answer Key : Part 1

Set-up

- Show Handout 1: **“Lawn Co. Case Overview”**
 - Other “Handout 1’s” are available if the interviewee requests the data version of what is prevented in the “Overview” slide

Key questions

- What is a good way to approach this question?

Model answer

- Identify that customer “profitability” (lifetime value) is key to understanding which growth option is better. Strong interviewees should lay out a profit tree-like framework
- Some interviewees may first ask about how fragmented the market is to assess the feasibility of pursuing an acquisitive strategy

-
- Show Handout 5: **“New customer counts and acquisition costs”**

- How much does it cost to acquire one new customer organically?
- How much does it cost to acquire on new customer inorganically?

- Cost to acquire in both cases is total acquisition costs / number of new customers acquired
 - Interviewees should round both the number of customers and total costs to numbers easy to work with
- Reasonable answers are \$360 for inorganic customers (\$18M / 50K) and \$100 for organic customers (\$100M / 1M)

-
- Show Handout 5: **“New customer counts and acquisition costs”**

- Why might inorganic customers cost more to acquire?

- The best candidates here might discuss purchase premiums or customer attrition during the sale process (e.g. don’t want to be a Lawn Co. customer)

Source: Bain & Co.

Interviewer Answer Key : Part 2

Set-up	Key questions	Model answer
• Show Handout 6: “Customer annual spend” and Handout 7: “Customer lifetime”	• How much does each type of customer spend over the course of their tenure?	• Total spend in both cases is average spend per year * total lifetime - Interviewees should round all numbers to numbers easy to work with • Reasonable answers are \$800 for organic customers ($\\$400 * 2$ years) and \$760 for inorganic customers ($\\$380 * 2$ years)
• Answer and wrap-up	• Given the information you know, which type of customers are better for Lawn Co.?	• Interviewees should calculate the “profit” or “contribution” for each customer type by subtracting the acquisition costs they calculated from the potential revenue - Yields \$700 for organic customers and \$360 for inorganic customers • The strongest candidates here might first ask if the cost to serve both types of customers is the same to see if subtracting the two values gives you a good proxy for relative attractiveness • Interviewees should identify organically acquired customers as more attractive

Credit Card Processor

Bain, Round 1

Situation

Our client is a portfolio firm, a credit card (CC) processor with two sets of clients

- Corporate (e.g. Papa John's)
- Small and Medium Businesses (e.g. Joe's Pizza Shack)

Complication

Client profitability and revenue has been decreasing for the past few years even though market has been growing

Key Question

We have been hired to determine the reason for the decline and to recommend ways to increase profitability

Difficulty: Medium

Graph Heavy

Industry: Financial Services
Profitability

Type:

Guidance for interviewer and information provided upon request

How does a credit card processor work?

- Candidate should at least talk about how the industry works, but provide following information if asked
- Processor doesn't provide CCs or handle cash debit/credit; it only enables the transactions by processing them when swiped on CC machines (point of sale)
- Client has large sales force that is in the field selling to both Corporate and Small and Medium Businesses (SMB)
- Typically 2 ways to make money
 - Leasing fees of equipment
 - \$ or % per transaction
- Our client only makes money with a fixed \$ per transaction (not from leasing or %)

Solution Guide

- Interviewee should draw a customized profitability framework but should include some of these important items.
- Price: Different pricing for corporate vs. SMB (Small & Medium Businesses)
- Quantity: Market size, market demand, growth in online vs. stable or reduction in offline due to recession
- Fixed Cost: IT system, overhead
- Variable Cost: Sales team
- Context: There is shift from in-store retail to online, could explain drop in client revenue and profit. Intense competition. Recession. Square, PayPal.

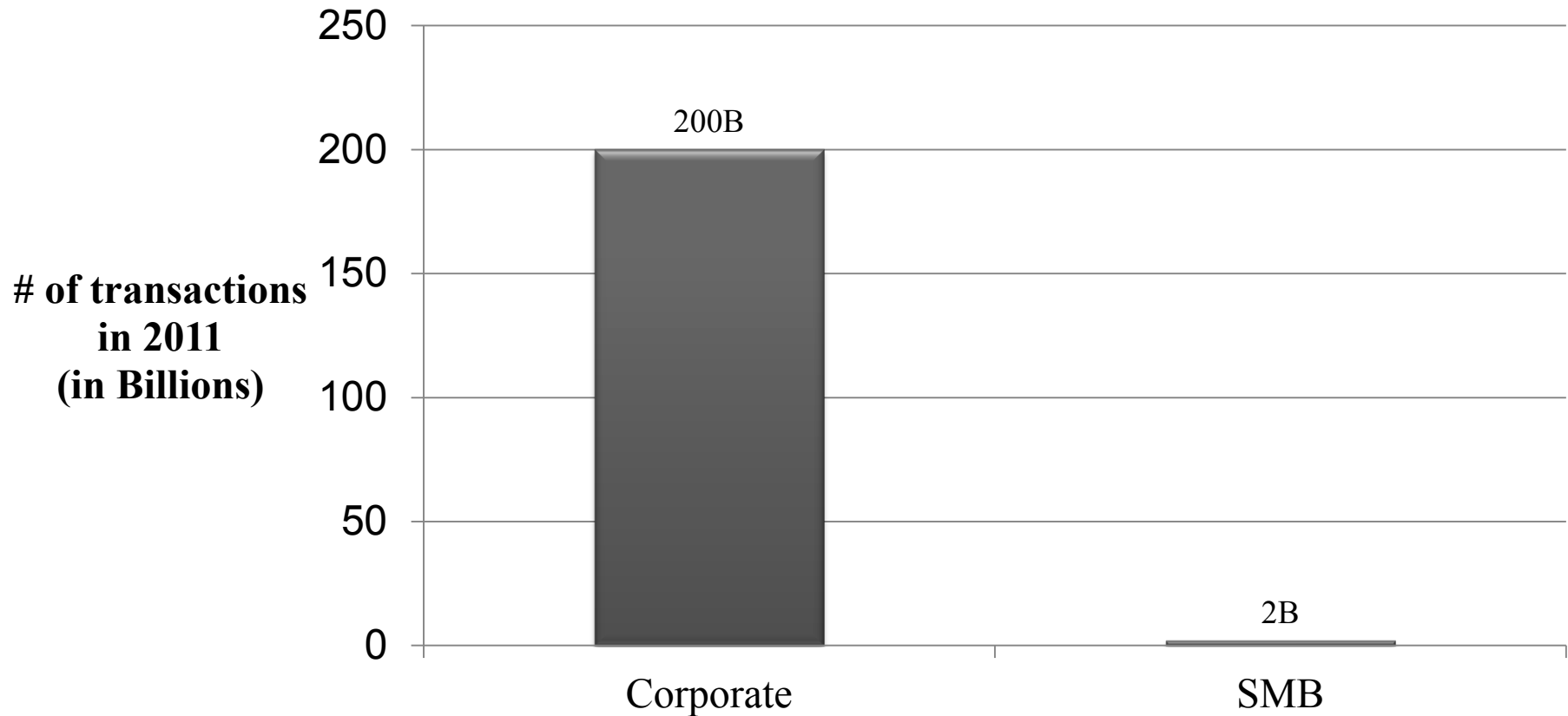
Additional Questions to Steer Discussion

Questions for the candidate

- Show exhibits to interviewee and then ask for trends (non-obvious trends highlighted):
 - Exhibit 1: Even though SMB sector is growing fast, it's still a very small % of Corporate
 - Exhibit 2: Should mention that in 2012/2013 cost > revenue for client, resulting in losses. Why client costs not going down ? No economies of scale, high sales cost
 - Exhibit 3: Considerably higher margin compared to Corporate – why? Bargaining power of big clients reduces price in that sector but SMBs don't have choice
- Ask to calculate profit for each sector: \$30M for both. Interviewee should have an “Aha!” moment and say this is the segment to focus on to grow profits
- Then ask for what to do with corporate client where we would start losing money. Offer information that we are stuck in multi-year contracts where clients only pay industry average, hence we start losing money on each transaction starting 2012/2013. What do we do?
 - Get rid of corporate clients, buy out most expensive clients, spin-off bad business unit, like GM, and declare managed bankruptcy, outsource, sell division, renegotiate, acquire lower cost competition

Exhibit 1:

Total # of Credit Card Transactions



YoY Industry Growth:

2%

10%

Client Market Share:

0.5%

5%

Exhibit 2: Cost vs. Revenue for Corporate Customers

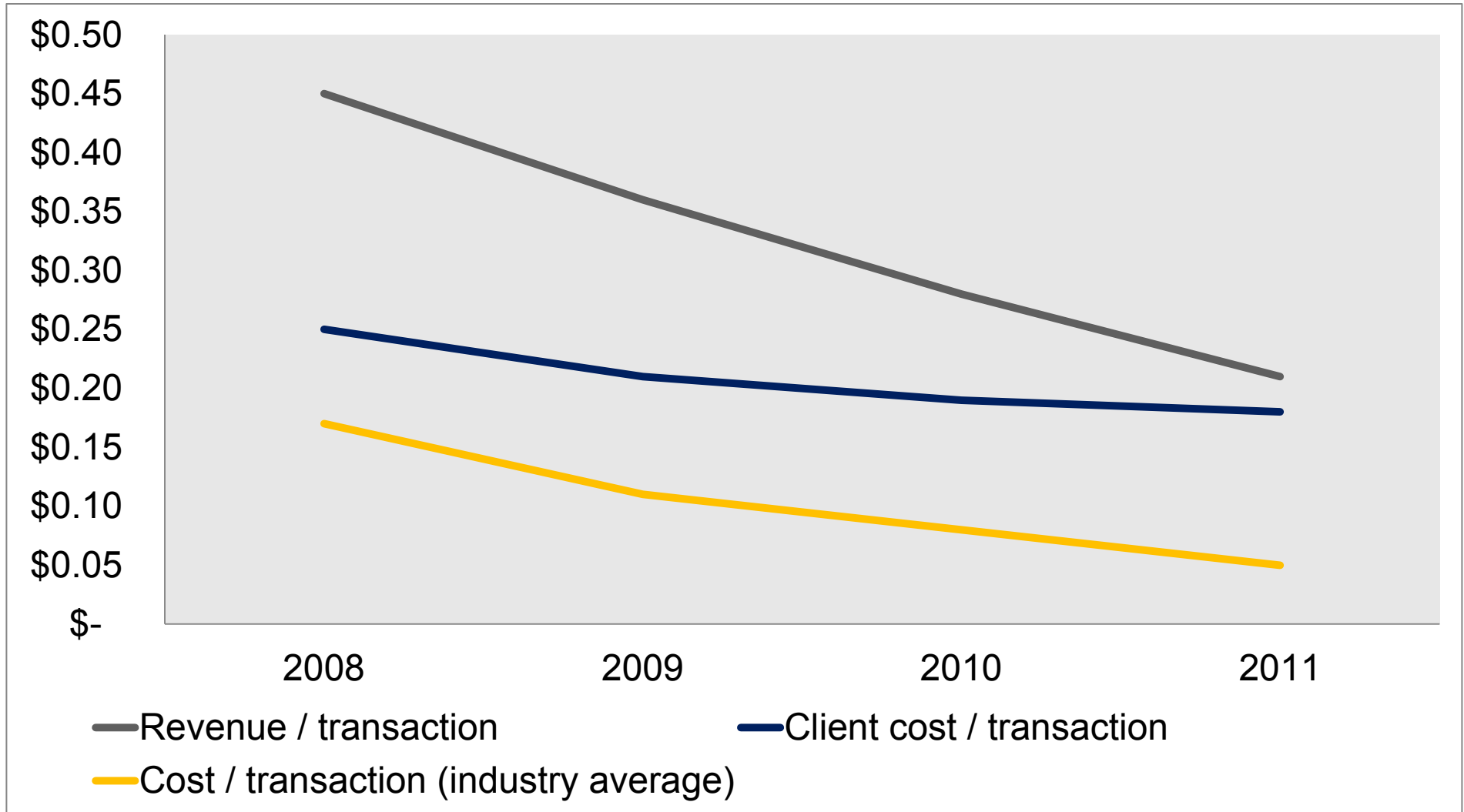
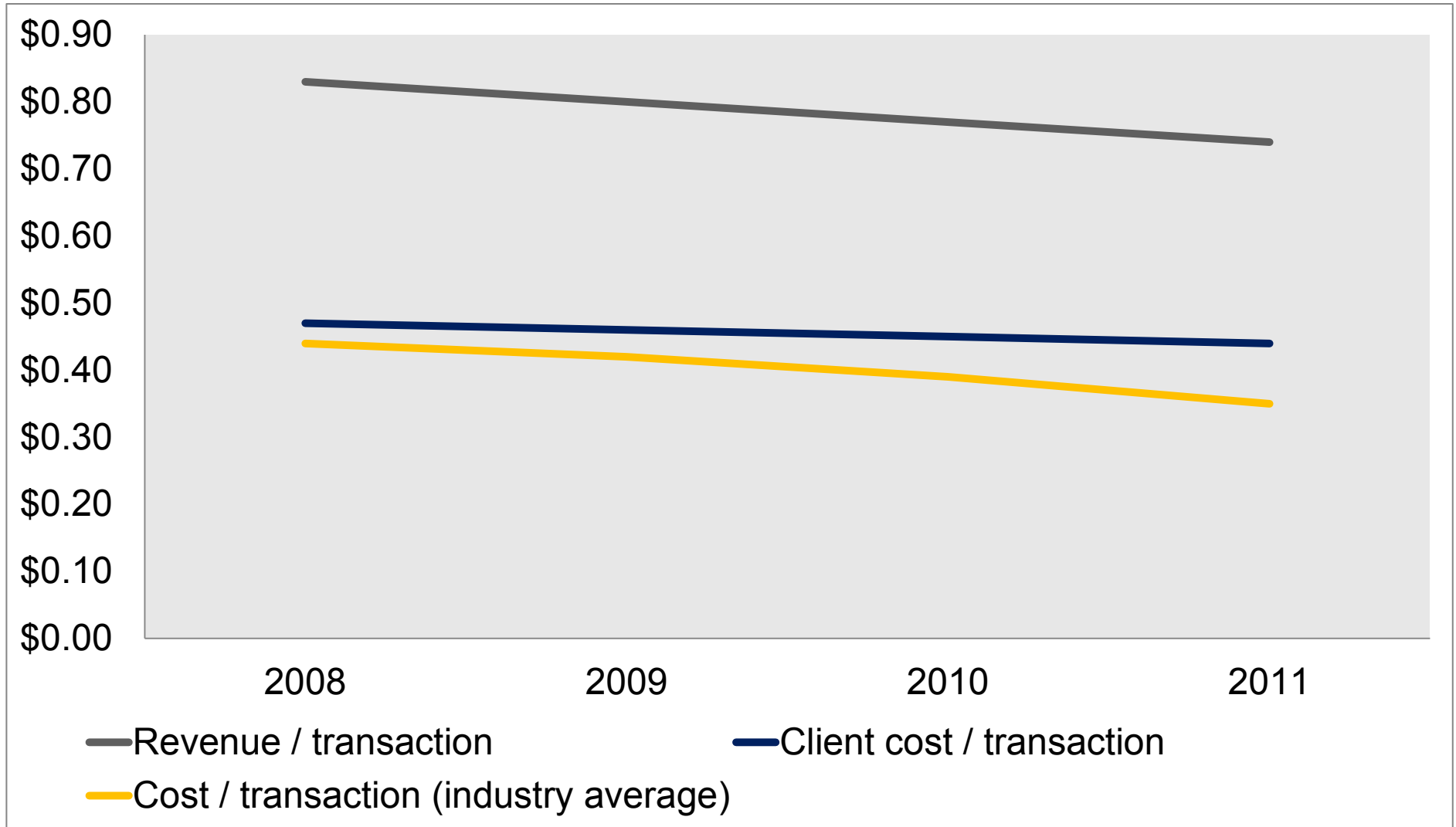


Exhibit 3:

Cost vs. Revenue for Small & Medium Business Customers



Conclusion

Recommendation

- **Drop Corporate customers to prevent losses from overwhelming and bankrupting client**
- **Reduce cost of sales by moving to telesales, online, or lower cost model**
- **Build relationships with banks for debit card processing for new growth**

Next Steps

- **Finalize steps to drop Corporate customers**
- **Overhaul sales model and start socializing new plan**
 - **Move sales team from salaried to independent contractors**
- **Invest in SMB segment**

Risks

- **Inability to drop/renege customers**
- **No/negative growth in non-online retail**

Little Bud Co - Beer Company

Bain, Final Round

Situation

Our client, Little Bud Co, is a beer company in a small country in Latin America. Little Bud and its main competitor, Geineken, are the only players in the market. Geineken is significantly bigger than Little Bud.

Key Question

Little Bud's CEO asked us to provide him with strategic options for the company and a recommendation on what he should do.

Comment

If interviewee asks "What are the CEO goals?", turn the question back: What are the goals of a company? Then rapidly lead conversation to the goal of maximizing shareholder's value

Difficulty: Medium

Quant Heavy

Industry: Food + Beverages

Type: Growth Strategy

Guidance for interviewer and information provided upon request

- **This case is focused on a discussion on how scale economies create competitive advantage and will touch on valuation principles at the end**
- **It's important for the candidate to rapidly start comparing the two companies**
- **For the purpose of this case, other competitors can be disregarded. The analysis should be focused on Little Bud and Geineken**
- **Market characteristics:**
 - **Market is mature (growth is low)**
 - **10% of sales are made through large retailers and 90% through bars, restaurants and small retailers**
 - **Market of regular beer: market for small brewers/specialty beers should be disregarded (very small market)**

Questions for the candidate

- **At the beginning of the case (after candidate presents his framework):**
 - **What are the two most relevant information that you need to start this analysis? Suggested answer: market size and P&L for Little Bud and main competitor**
 - **Once the candidate asks for market size data, present exhibit 1 and ask for initial insights**
- **After presenting exhibit 1:**
 - **What are the margins for each player? What are the possible reasons for the margin differences? (suggested answer: Geineken could have higher margins due to higher prices, lower costs or both). When candidate raises the price hypothesis, hand over exhibit 3.**
 - **What are the strategies that Little Bud should consider? (suggested answer: go to niche market, go to related markets, sell to competitor)**
- **After analyzing both companies (refer to next page for more details on this analysis), the interviewer should focus on discussing the company valuation**

Calculation and Discussion: Comparative Analysis

Comparative Analysis of Little Bud and Geineken

At this point, the candidate should try to compare the two players, in order to identify the possible sources of competitive advantage. The table below presents information on Geineken costs as a percentage of sales (so the candidate can calculate the P&L) and the rational behind the cost differences between Little Bud and Geineken. The interviewer should ask the candidate to provide hypothesis on the rational before explaining Geineken's competitive advantages

P&L Item	As % of Revenues	Rational
G&A	10%	The assumption here is that Little Bud already operates in a scale in which there are no more scale economies in G&A. The candidate might make the valid argument that "the bigger, the better", but should recognize that scale economies have a limit
Sales & Distribution	25%	Info: Due to market regulations, there is a limit in the truck size. Both companies operate with similar trucks and sell through the same point of sales (or through point of sales located in the same regions) Rational: The analysis here is about utilization: due to its larger share, Geineken achieves higher asset utilization (consider one truck: at Little Bud, this truck would take a whole day to delivery the products, given the low number of units per location. At Geineken, this same truck can deliver the full load faster and go back to the distribution center, pick up more products and get back to the road)
Marketing	18.1%	Rational: even with a lower percentage, the marketing expenses are much higher in absolute terms. More advertising gives more brand recognition, reinforcing higher market shares

Calculations and Discussion:

Conclusion of Comparative P&L

Options for Little Bud Business

- After comparing both companies, the candidate should understand that scale economies are a major source of competitive advantage in this market and propose some strategic alternatives to Little Bud. The interviewer should rapidly move the discussion towards selling the company. Below are some arguments against other possible alternatives:
- Increase market share:
 - Product innovation: consumer taste is already well defined. There is no room for new products
 - Price war: it's virtually impossible for Little Bud to enter a price war against Geineken. It's more likely for the opposite to happen
 - Marketing: Geineken spends 5x more in marketing. Trying to compete against this big player on marketing expenditures is not a good strategy
- Reduce costs: interviewer should state that different assessments on cost reductions have already been conducted. There are no opportunities for further reducing costs
- Focus on niche market (e.g.: premium beers): interviewer should remind candidate that niche market for beer in this country is very small. It's not a good strategy for sustainable long term growth
- Move into related markets (e.g.: carbonated soda, juices, etc.): interviewer should say that the company has already conducted analysis on these markets and decided it's not a good alternative, due to intense competition

Calculations and Discussion: Valuation

Valuation of Little Bud Business

- The interviewer should propose to focus on the valuation of the business and ask the candidate to propose how to value the company with the information given in the case
- The candidate should be able to propose at least the two most common valuation methods: discounted cash flows and multiples: multiples are used for quick assessments and we have EBITDA figures. Once asked for it, the interviewer should give the multiple to be used: $EV/EBITDA = 10x$
- Going back to our initial questions regarding valuation. The goal now is to test the candidate's business sense:
 - What is the market value for Little Bud? Suggested answer: based on the multiple and EBITDA figures, \$50M
 - What is Little Bud's value to Geineken? Suggested answer: considering Geineken margins and Little Bud revenues, Little Bud's value to its competitor is \$214 (potentially higher, due to the fact that Geineken would become a monopoly and be able to increase price and further squeeze suppliers)
 - How to "force" Geineken to pay more than \$214? Suggested answer: Little Bud should open bid to other companies. Geineken has the incentive to maintain its market dominance and doesn't want a big international player to enter the market

Calculation for Comparative P&L

P&L Calculations						
P&L Comparison					Pricing	
	Little Bud	% of Rev	Geineken	% of Rev		
Revenues	100		700		Unit Price	1,00 1,10
COGS	30	30,0%	178	25,5%	COGS	0,30 0,28
Gross Profit	70	70,0%	522	74,5%	Gross Profit	70% 75%
G&A	10	10,0%	70	10,0%		
Sales & Distribution	30	30,0%	175	25,0%		
Marketing	25	25,0%	127	18,1%		
EBITDA	5	5,0%	150	21,4%		

Conclusion

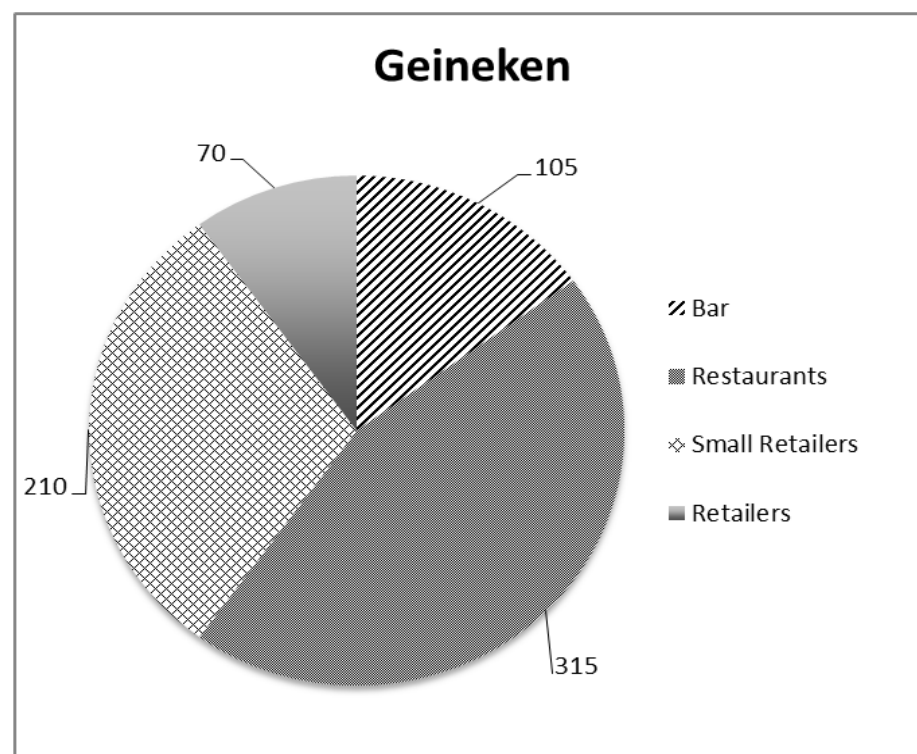
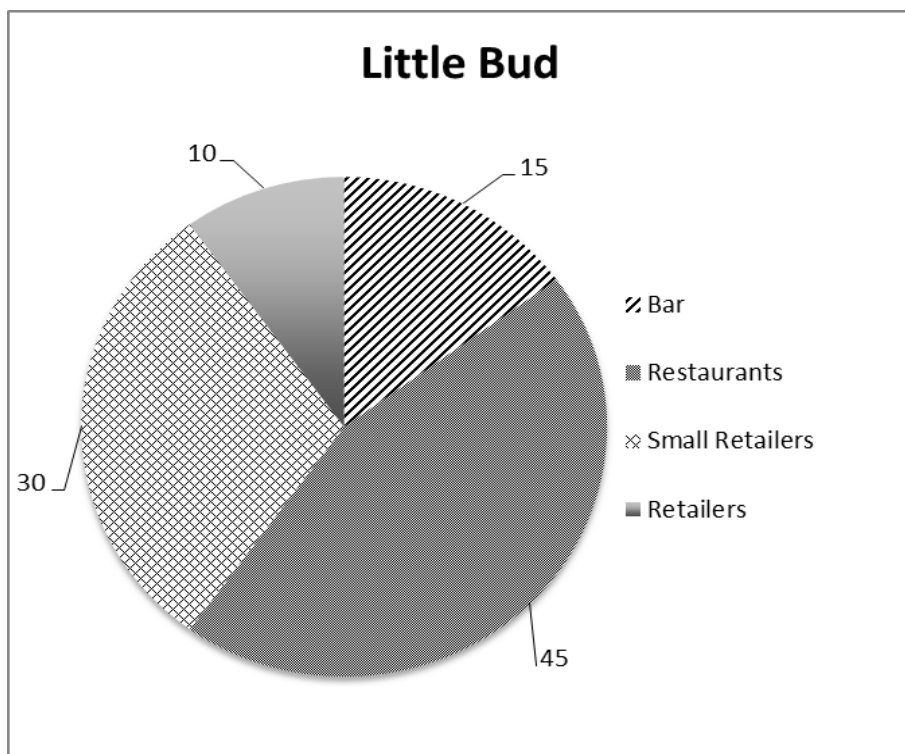
Recommendation

- **The candidate should recommend the CEO to sell Little Bud**
 - **The company currently operates in a market in which it has no competitive advantage. Geineken has margins high enough to enter a price war that would lead Little Bud to bankruptcy.**
- **Also, through an open bid to the market, Little Bud could force its main target buyer to increase a potential offer**

Next Steps

- **Hire an investment bank to structure the deal**
- **Evaluate risk of ruling from antitrust agencies**
- **Identify potential cost savings opportunities that could be pursued in case Geineken responds by reducing prices**

Exhibit 1: Revenues per Channel



Figures in USD millions

Exhibit 2: P&L for Little Bud and Geineken

P&L Comparison		
	Little Bud	Geineken
Revenues	100	700
COGS	30	
Gross Profit	70	
G&A	10	
Sales & Distribution	30	
Marketing	25	
EBITDA	5	150

Exhibit 3: Gross Profit Detail

Average Price and Cost Per Unit		
	Little Bud	Geineken
Unit Price	1,00	1,10
COGS	0,30	0,28
Gross Profit		

Digital Expansion Prescott Publishing

Original case and solution from Booz & Co.

Business Problem Statement

Your client is Prescott Publishing, a major business media company. They have a wide range of product offerings, including general business magazines, industry trade letters and branded television programming. Performance has suffered recently, as print circulation has declined. The client leadership team wants to reboot their offerings with a new premium digital product. It wants to specifically target business executives with a paid web news offering. They would like your help to understand the opportunity and whether you would recommend they go forward with an investment?

Guidance for interviewer and information provided upon request

- Company is focused solely on the U.S. market
- Company is focused on a subscription model, any other revenue could be explored but will initially be immaterial
- Opportunity should elicit a negative recommendation.
- Given time, push the candidate to identify ways to improve the concept
- Framework should include:
 - Market Size/Revenue Opportunity
 - Potential Customers
 - Subscription Pricing and Other Revenue
 - Cost (infrastructure, talent, etc.)
 - Could also include Competition, Risks, etc.

Guidance for interviewer and information provided upon request

Additional Question

What kind of product is it?

Do they know what price they want to charge?

What does the company think defines ‘an executive’?

What does the company expect it will cost?

Guidance for Interviewer

The client wants to become the ‘Home Page’ of the modern executive, providing them with current insights and trends on their industry, deep subject matter expertise, and relevant data.

Give the candidate competitive exhibit and ask their opinion. They should want to focus between the \$75 and \$350 options given the product description. After they make an estimate, tell them the client thinks customers would pay \$10/month

To the client, an executive is either someone in the top five leadership positions (CXOs), one of the top 6 direct reports to those positions, or one of their 6 direct reports (5x6x6=180 per company). They’ll also typically work for a large public company.

Push the candidate to identify different buckets of costs for such a product (e.g., infrastructure, labor costs, marketing, etc.) The company anticipates the construction and launch of the product would cost up to \$30M, and would remain that significant going forward

Exhibit 1:

Competitive Landscape

Competitor	Pricing Model	Offering Description
Wahoo	Free	– Company financial reports – General Business News
Froogle	Free	– SEC filings – Company news
Wealth Publishing	\$24.99/year	– Executive-focused lifestyle articles and general news
Lews Corp	\$74.99/year	– In-depth business articles – Industry Analysis
Woodson Research	\$350.00/year	– Customized research reports on specific industries/companies

Sample candidate response

Sample Solution

I'd like to start by clarifying exactly what the product is so we can then size its potential market and evaluate whether we'll pursue the opportunity

The product will be the executive's home page

- Provides deep industry insights and knowledge
- Analysis of breaking events impacting their company
- Provides equivalent to a 'Morning Briefing', all the critical information an executive needs for the day
- Premium, will focus on revenue through paid subscriptions, little if any emphasis on advertising

Now that we know a little more about the product and its revenue model, we can develop estimates of what it can generate

- Based on competitive landscape and company research (*to be provided*) a monthly subscription price of \$10 seems reasonable, of \$120 annually
- Based on the client's target market, we can estimate that there are 180,000 people in the target audience (*As noted in next pages: 5x6x6=180 per company, x Fortune 1000, 180,000*)
- From these two estimates we can establish a potential market opportunity of about \$22M
- However, we should also consider that people outside the target market may be likely subscribers as well

Sample candidate response

Sample Solution

With a \$22M potential market size, we should now turn our attention to the investment's cost

- Costs will include infrastructure for a new web site, design, technical and reporting staff, maintenance, and marketing
- The initial estimate of costs places the new site at \$30M annually

At that projected cost, the new product will require many more subscribers than our target market. This is before accounting for any competitive risks. As such, we would not recommend investment.

To make the investment case more attractive, there are several potential levers that could make it a stronger opportunity

- Mobile App complements
- Higher pricing (for a premium product)
- Broader targeting (small and medium size businesses)
- Advertising (keeping in mind a premium position)
- Providing additional value beyond media (e.g., networking, industry events)

Sample calculations

Revenue/Customer - $\$10/\text{Month} \times 12 \text{ Months} = \$120/\text{customer}/\text{year}$

Customers – $(5 \times 6 \times 6) = 180 \times 1000$ (Number of Companies) = 180,000

Total Target Market Size = $\$120 \times 180,000 = \21.6M per year

OR

Breakeven customer number = $\$30\text{M}/120 = 250\text{k}$ customers (ask how many companies you'd need to target),
 $250\text{k}/180 = \sim 1400$ companies

What a strong interviewee should consider

Additional revenue through advertising, mobile apps, other creative ideas

Opportunity cost of investing in this product

Risk of new competition providing information for free

Adoption of product by non-executives (i.e., those outside the target audience)

V-Formulae Overview

BCG Round 1

Situation

Our client “V-Formulae” is a baby food formulae producer. The client’s main ingredient in most of the products that it sells is milk. The client has machinery that pasteurizes the milk and evaporates it to create the powder, which is then used in most V-FORMULAE products available in the market.

Complication

The client’s profits have declined over the last three quarters, and the client suspects that the cost of milk is affecting the margins.

Key Question

A supplier approached our client and has offered a five year fixed contract to supply milk at 10% below the current market price. We need to help our client figure out if they should enter into this contract with the supplier.

Difficulty: Medium

Industry: Food + Beverages

Type: Profitability, Supply + Demand equilibrium

Interviewer Guide: Case Overview

Background

- Industry – CPG
- Concepts tested – Ability to thinking laterally and macroeconomic concepts
- Level of Difficulty – Hard

High-Level Model Answer

The candidate should be given time to brainstorm and come up with an approach. A good candidate will immediately grasp, that the problem is to figure out if the market prices of milk will remain constant, increase or decrease over the five year period.

A good structure would focus and touch on the following points:

- Reason for Increase in cost of milk
- Supply vs. Demand of milk
- The type of milk the company uses (goat, cow etc.) other alternatives
- Sourcing strategy for milk
- Possibility for backward Integration into milk production
- Historical prices for milk
- Realizing that milk is a commodity and that prices will be dictated by the market
- Recognizing that if we do not sign the contract our competitors may sign up the contract
- Demand in the emerging economies driving up the prices for milk

Additional Information

Our client is US based and all of its sales are only in the US.

We are the market leaders however there are two other major competitors who have a significant market share.

There are numerous smaller players but they do not have more than 10% of the market.

Interviewer Answer Key : Step 1

Set-up

- Allow the interviewee to think broadly and come up with major macroeconomic issues
- Provide the information on cost structure and margins Exhibit 1

Model answer

- The average price of end product has more or less remained stable indicating this is price elastic and V-FORMULAE does not have a lot of leeway with increasing the prices
- The price for milk has increased by 28.5% over the last five years.

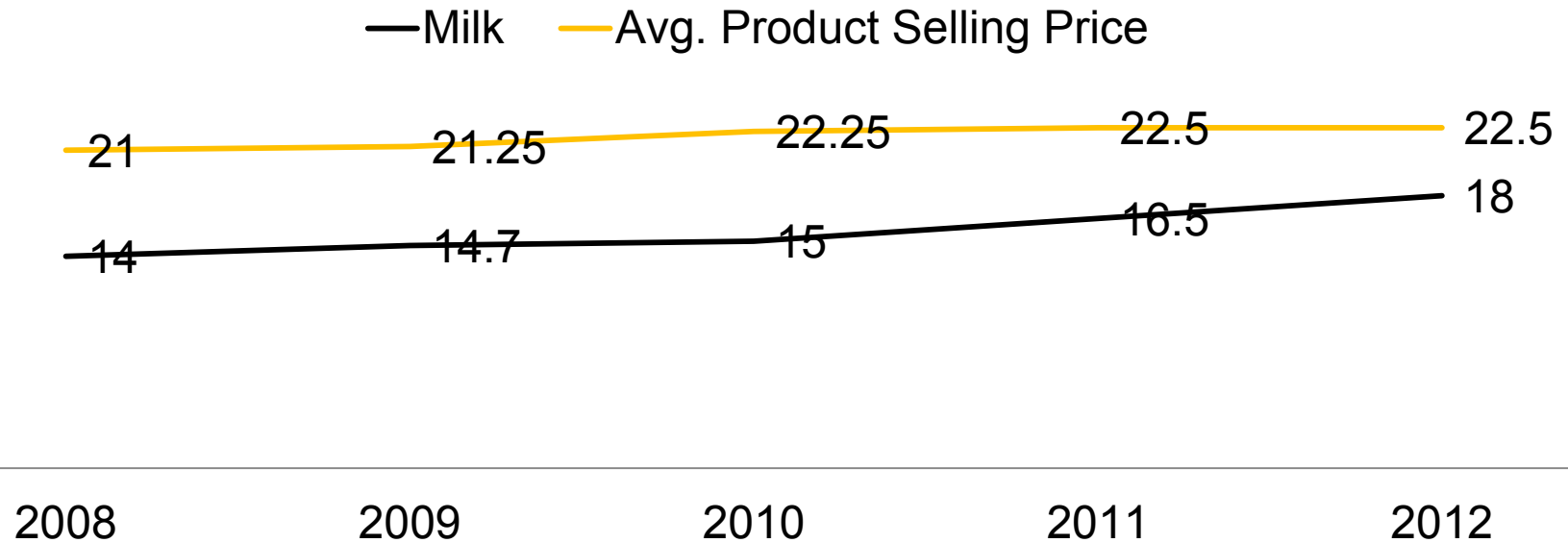
After this table the candidate gets brownie points if they bring up reasons on why the price has gone up. The next step for the interviewer is to focus on reasons why this price increase has happened. Brainstorm with the candidate to figure out the major reasons for the increase in price of the milk. A good candidate will clarify if the chart showed above was only for a certain kind of milk e.g. goat vs. other. In which case clarify that our company only sources Cow's milk today.

A good candidate will touch upon the following points while discussing the major reasons for increase in price of the milk

- Increase in demand in the global market because of demand in emerging economies
- Supply side difficulty e.g. draught causing livestock maintenance prices to increase
- Analysis of who are the major producing states for milk and if there are opportunities to import it
- Output / cow and possible reasons for its decline? Health issues affecting output
- Livestock feed costs increasing.
- Decrease in farmland and hence number of cows producing milk

Exhibit 1

	2008	2009	2010	2011	2012
Milk Price	14	14.7	15	16.5	18
Average Product Selling price	21	21.25	22.25	22.50	22.50



Interviewer Answer Key : Step 2

Set-up

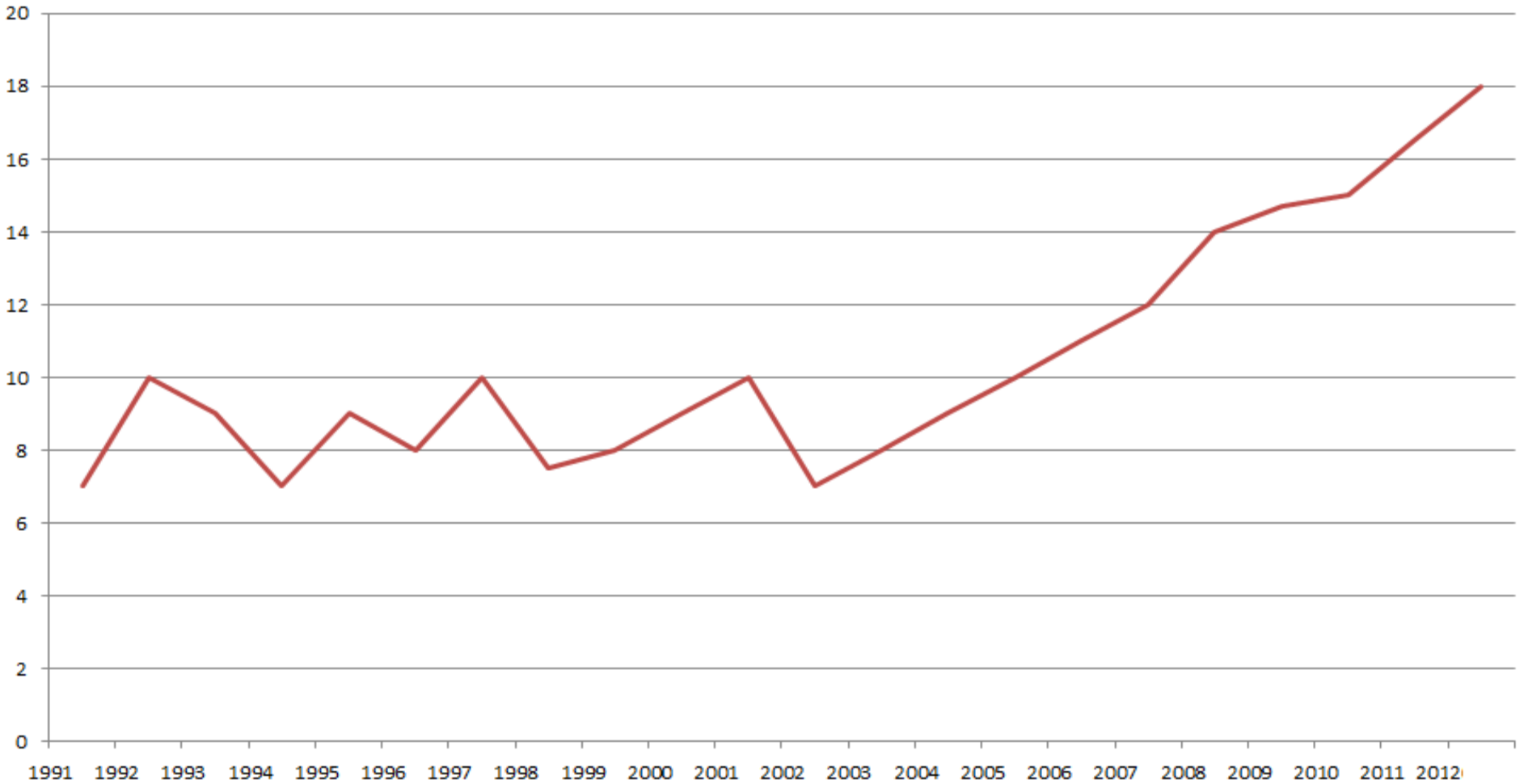
The interviewer should at this point present Exhibit 2 to the candidate

Model answer

- The milk prices crossed the historical highs in 2005 and since then have been rising continuously
- The candidate should ideally talk about their hypothesis on why the rates shot up so high. The demand in the global market that would be a good point to explain the rise of prices
- After spending a few minutes and letting the candidate brainstorm and talk about this chart

Exhibit 2

Avg Price/Gallon-Milk



Interviewer Answer Key : Step 3

Set-up

The interviewer should at this point present Exhibit 3 to the candidate

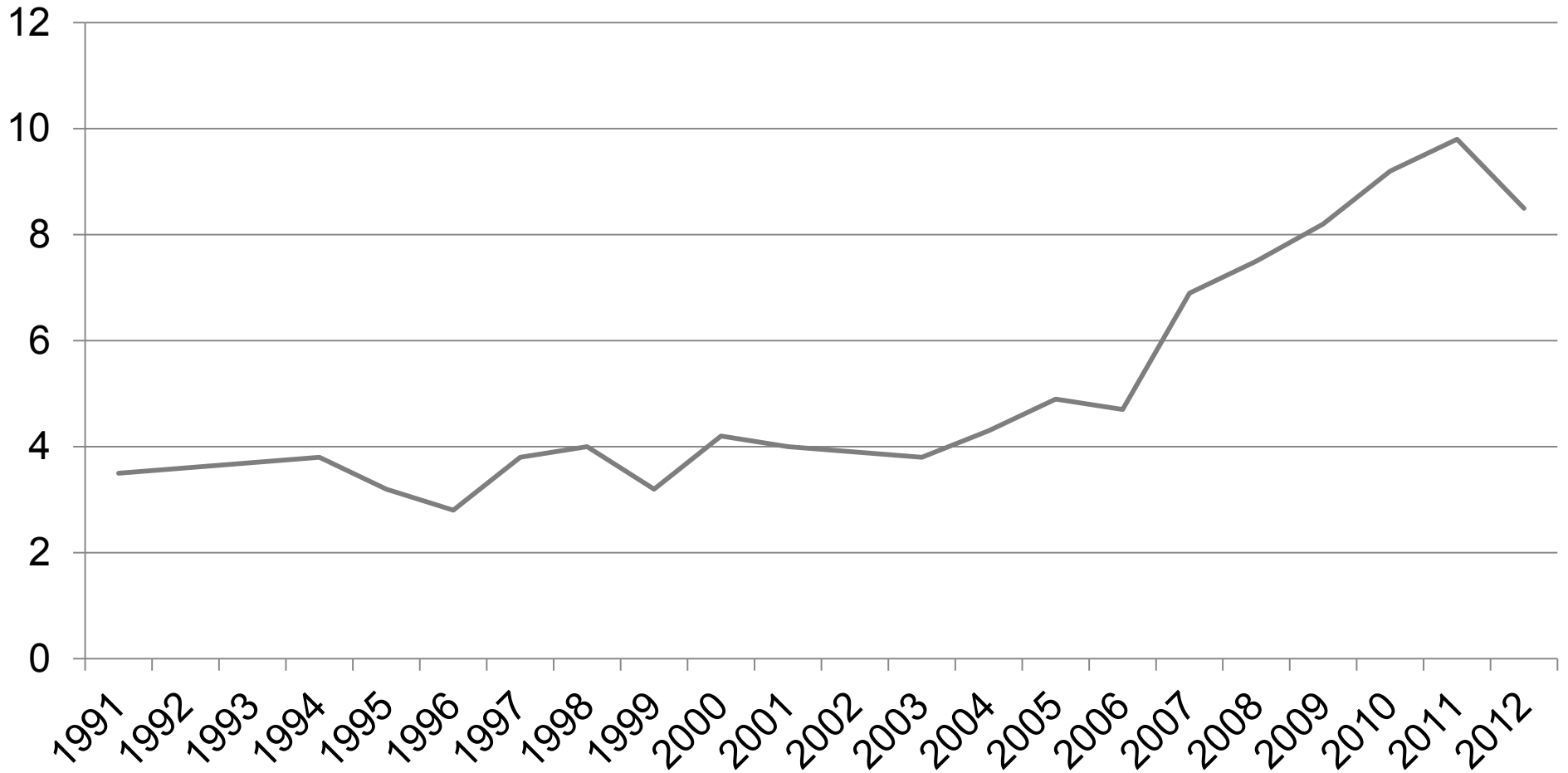
Model answer

- Clarify if the candidate asks that corn is the major livestock feed used across all cow farms and the chart indicate its historical prices
- A good candidate infers that it has been the increase in corn prices that has led to increase in the price for milk
- Interviewer Guidance
- A government mandate to use Corn for Ethanol in 2008 drove up the prices of corn across the board.

Direction for interviewer: The government mandate expires in 2012 and the prices of corn are expected to fall

Exhibit 3

Price of corn/Bushel



Conclusion

Recommendation/Next steps

The current offer price for the five year contract is \$16.20, while the terms are attractive the falling livestock feed prices indicates that the milk prices will come down sooner than later

The client could explore the possibility of using alternate sources of milk like goat milk – provided the nutrient content level is the same and it does not pose any nutritional hazard

Five year is a very long time horizon and probably a 1- 2 year time horizon would make more sense to enter the contract

The client should work very closely with the milk farms to understand their cost structure and help them reduce cost by possibly using alternative feeds like soybean

Retail Co. HR Structure

Accenture Talent & Organization (Human Capital), 1st Round

Problem statement narrative

Background:

From 2007-2009, Retail Co. had the largest revenue (\$400B) in the world. For the purposes of this case, let's just focus on U.S. operations. In the U.S., Retail Co. has 1.3M employees who work in 4100 facilities across every state. These facilities include Super Centers, Neighborhood markets, regular stores, Sam's Clubs, distribution centers, and corporate. Retail Co.'s strategy is to be the low price leader, and it has core competencies in supply chain, distribution, manufacturing, analytics, and negotiating low costs with suppliers.

Case:

Retail Co.'s HR department structure has not changed much since the 1970s. HR is not organized well as compared to Retail Co.'s other departments. The typical cost in a large corporation to provide HR services should be \$1500-1700/employee/year, but at Retail Co. the cost is currently \$2100-2200/employee/year. Additionally, a corporation of Retail Co.'s size generally has a 1:100 ratio of HR: employee support, but at Retail Co. the ratio is 1:33. The CEO has hired you to provide recommendations as to how to lower HR costs and increase efficiencies. Specifically, he is looking for ways to decrease HR operational expenses while increasing overall revenues.

Difficulty: Medium

Industry: Retail

Type: HR/Admin

Interview Guidance

Guidance for interviewer and information provided upon request

- What are the causes of high HR costs?
 - Retail Co. 's HR operations are very labor-intensive. Very little automation or outsourcing (e.g., Fidelity handling 401K); almost everything is handled in-house.
- Does Retail Co. utilize HR technology for support?
 - Retail Co. is currently not using technology, but the CEO has indicated that he is interested in exploring this option.
- What is the current structure of the HR organization?
 - There are currently 7K full-time HR professionals (both corporate and regional). Additionally, there is an additional 26K FTE HR work being completed by 104K store managers (in ~25% of their time). The CEO is especially concerned about the 26K FTE because this has opened up Retail Co. to compliance risks since these store managers are not skilled or certified HR professionals.
- What is the salary for the HR professionals?
 - HR professionals make ~\$40K/year
 - Store managers make ~\$100K/year
- Managers need to spend at least 12.5% of their time on HR related activities

Discussion Questions and Solution

Solution

Reduce manager HR workload and reallocate work to HR specialists

Current HR costs:

- $7K * \$40K = \$280M$
- $26k * \$100K = \$2.6B$
- Total: \$2.88B

HR Costs with work transferred from store managers to HR professionals

- $(7K + 13K) * \$40K = \$800M$
- $13K * \$100K = \$1.3B$
- Total: \$2.1B
- HR cost per employee: $\$2.1B / 1.3M \text{ employees} = \$1600/\text{employee}$

Candidate should also consider:

- Activities managers can do with freed time
- Outsourcing options
- Shared-service technology options and capabilities
- Compliance issues of having store managers work on HR activities

Conclusion

Recommendation

- Increase the number of HR specialists and free managers to focus on selling
 - Approximately \$800M in cost savings
 - Reduced liability of store managers performing HR tasks
 - New HR cost per employee consistent with industry standards
- Risks
 - Managers need to demonstrate value with additional free time, otherwise the change cannot be justified

Next Steps

- Increased use of technology to manage HR activities between stores
- Evaluate outsourcing options for new HR positions
- Adjust metrics for manager performance

Melson Medical Scanner

McKinsey, Final Round

Problem statement narrative

Your client, Melson, is a dominant (98% market share) manufacturer and seller of CT scanners in the US hospital market. The market is quite stable, and the U.S. demand for images from a CT scan grows 5% every year. At the beginning of last year, Melson launched a new CT scanner model that is sold at the same price but can produce 20% more images per year than the previous model. However, apart from the sales due to product replacement, the sales growth of the new model has been stagnant in the last year.

The CEO has hired McKinsey to find out the reason for stagnant sales growth, and asked you how to increase sales.

Guidance for interviewer and information provided upon request

This is a interviewer-led case.

- CT scanners produce images of patients for doctor to diagnose disease.
- Melson only manufactures and sells one model at any time. i.e., when the new model goes to market, they stop manufacturing and selling old models.
- The life time of old and new model are both 8 years.
- Hospitals match their number of scanners with the image demand. There is no idle capacity, and new purchase will be driven by either replacement or increasing demand.

Difficulty: Medium

Industry: Retail

Type: Growth strategy

Discussion Questions & Solution

Questions for the candidate

- Q1. Why has the sales growth of the new model been stagnant?

Solution Guide

- The image demand of CT scanning in the US only grows 5% every year, while the new model can produce 20% more images than the previous model.
- **Conclusion**: by introducing a new model, the total demand for CT scanner will actually go down!

Discussion Questions & Solution

Questions for the candidate

- Q2. What is the new sales of the new model (i.e. total sales excluding replacement sales)
- Note: when interviewee asks about the revenue, sales price and volume, push her/him back to let her/him assume some figures and figure out what can be calculated!

Solution Guide

- Before the new model launch (the year before last year):
 - Let the total number of existing old-model scanners in all hospitals be X
 - Let the number of images produced per old-model scanner per year be S
- In the last year:
 - The replacement sales will be $X \cdot (1/8)$
 - New model can produce $1.2S$ images per scanner
 - Let the new sales of new model be Y
- Last year's image demand: $X \cdot S \cdot (1+5\%)$
- Last year's image production:
 - From old models: $X \cdot (7/8) \cdot S$
 - From replacement (using new model): $X \cdot (1/8) \cdot 1.2S$
 - From new sales of new model: $Y \cdot 1.2S$
- Demand = Production $\Rightarrow X \cdot S \cdot (1+5\%) = X \cdot (7/8) \cdot S + X \cdot (1/8) \cdot 1.2S + Y \cdot 1.2S \Rightarrow Y = 2\%X$
- **Conclusion**: excl. replacement, the new sales of the new model is only 2% of existing old-model scanners

Discussion Questions & Solution

Questions for the candidate

- Q3. How to increase sales
- Note: This is brainstorming, so keep asking “what else” until the interviewer runs out of ideas

Solution Guide

- Key to brainstorming is being structured. Creativity is always a great bonus!
- 1. Increase price
 - We are in a monopoly, and we offer a new model that is better than old model. So increase price!
- 2. Increase volume
 - Enter into new segments, e.g. new geographic regions, medical research center, clinic, etc.
 - Focus on improvement on image quality rather than number of image produced.
 - Shorten life time of scanner and educate hospitals on this. (by significantly improving quality, the old models will be obsolete and needs to be replaced before it is not functioning)
 - Increase demand of CT scanning: Partner with media, healthcare association, key opinion leader, etc, to educate patients and physicians that CT scanning is safe to use, and the benefit of diagnosing early-stage disease can save people from death.
 - Improve marketing & sales force.

Conclusion

Recommendation

- Sales growth is stagnant because replacement are more effective than previous models
- How can Melson increase sales
 - Increase price for new models – new product is superior and client has a monopoly
 - Enter into new segments, e.g. new geographic regions, medical research center, clinic, etc.
 - Focus on improvement on image quality rather than number of image produced.
- Risks
 - Replacement rates drop as a result of the price increase

Next Steps

- Opportunities to grow quantity sold
 - Improve marketing and sales force
 - Significantly improve quality, the old models will be obsolete and need to be replaced before it is not functioning
 - Increase demand of CT scanning: Partner with media, healthcare association, key opinion leader, etc., to educate patients and physicians that CT scanning is safe to use, and the benefit of diagnosing early-stage disease can save people from death.

Bolsen Pharma

Bain, Final Round

Problem statement narrative

Suppose you are a Bain Consultant. In 30 minutes, you are scheduled to meet with a Bain Partner and the CEO of a \$35B Pharma client - Bolsen, and present your project update. However, the CEO just called saying that he would like to change the topic of the meeting and asked the Partner one question: should the company enter the oncology (cancer) market? The CEO is eager to hear your ideas in 30 minutes.

Now you need to discuss with the Partner and prepare a recommendation.

Guidance for interviewer and information provided upon request

This is an interviewee-led case.

- Bolsen has mature products, high capitalization, but little growth, and confronts increased competition from generic drug manufacturers driving down prices.
- Bolsen will enter the oncology market if that new market can help Bolsen gain double-digit growth on its overall business.
- Bolsen currently has very basic R&D capability far below the required R&D level for oncology products.
- Bolsen operates in the US and is only concerned about US market.

Difficulty: Hard

Industry: Pharma

Type: Market Entry

Information for Candidate

Further information to be provided upon request

Interviewee is expected to know that US population is 320M and life expectancy is 80 years.

Market size:

- Cause of death: cancer, heart disease, diabetes, etc. Assume that cancer accounts for 25% of all deaths every year.
- On average, cancer patients can live for 5 years from the time they are diagnosed with cancer to the time of death
- In that 5 years, each cancer patient gets 2 treatments on average.
- Drug expense per treatment: \$18K

Competition:

- Concentrated. Strong player in the oncology market like Genentech has 5 great products and takes 35% market share.
- Basis of competition is product, technology.

Possible M&A target:

- All the strong market players including Genentech have been acquired by large Pharma, except for Selgen in New Jersey
- Selgen has 2 great products and 10% market share

Client capability

- Client has a lot of cash and access to capital market

Suggested Solution and Structure

Suggested Structure

- Oncology Market
 - Size
 - Growth
- Competition
 - Share & concentration of current players
 - Basis of competition
- Method of market entry
 - Realizing that Bolsen's basic R&D capability is far below the required R&D level for oncology products, interviewee should suggest M&A and check with client's financing capability
 - Screening of M&A target
- Client capability
 - Financing capability
 - M&A experience
 - Post-merger integration issue: culture, etc.

Suggested Solution and Structure

Suggested Solution

- Oncology Market
 - Size
 - Death per year: $320M/80=4M$
 - Death per year due to cancer= $4M*25\%=1M$
 - At any time, the number of people living with cancer= $1M*5=5M$
 - Treatment per year= $5M*2/5=2M$
 - Market size= $2M*18K=\$36B$, a huge market
 - Growth
 - Ask interviewee to think about the drivers for growth
 - Possible drivers: people take more frequent treatments and are more willing to pay; people live longer with cancer because of the treatment and early diagnose of cancer.
 - Conclusion: high growth rate (actually, over 20%)
- **Conclusion**: Attractive market. Market size is huge with high growth rate. If client can get 10% of the oncology market, they will get \$3-4B revenue to meet their double-digit revenue growth target.

Suggested Solution and Structure

Suggested Solution

- Competition
 - From the example of Genentech and basis of competition, we can infer : 1 great product roughly corresponds to 7% market share. ($35\%/5=7\%$)
- M&A
 - We need to look at players ideally with 2 great products to get a 10%+ market share in oncology
 - Selgen is an ideal target
- Client capability
 - Client has very basic R&D capability far below the required R&D level for oncology products.
 - Client has a lot of cash and access to capital market
- **Conclusion**: competition and client capability suggest that client pursue M&A. Selgen is an ideal target.

Suggested Solution and Structure

Additional Questions for the candidate

- We found that only Selgen meets the client's requirements and hasn't been acquired. But remember, all large Pharmas go through the same logic like ours. Then, why don't they buy Selgen?

Possible answers

- Selgen doesn't want to sell itself
- Selgen's R&D people have already left the company
- Selgen's patents will expire soon
- Selgen asks for very high price (this is the actual reason. They ask for X20 multiple, while normal case is X10 multiple)

Conclusion

Recommendation

- Acquire Selgen to enter the oncology market
 - \$36B large market with high growth rate
 - Client has no enough R&D capability but enough cash
 - Selgen is the last chance for us to enter oncology
 - With Selgen, client can gain double-digit growth
- Risks
 - Pay high price for Selgen
 - Integration issue
 - Past M&A experience

Next Steps

- Help client conduct a due diligence of Selgen to prepare M&A
- Find ways to boost revenue of Selgen by 100% ASAP, so that the actual multiple will be X10
- Client needs to be committed to succeed in the oncology market. Many large Pharmas executed acquisitions with no focus and poor results due to execution and integration issues

African Call Center

McKinsey, Round 1

Problem statement narrative

Your client is a large retail bank in the U.S. looking to move it's current outsourced call center from India to Africa and is currently evaluating 3 possible countries as a target location

How would you evaluate each of these sites?

Guidance for interviewer and information provided upon request

- Call center is focused on two types of calls:
 - Customer Service Calls (typical ones like Account locked, password reset, etc.)
 - Sales Calls (pushing new services to new and existing clients e.g. credit cards)
- This is the Bank's first time in Africa
- Candidate should recognize that this is a cost reduction case primarily
- Bonus: Candidate should identify inherent upfront risks of moving call center to brand new market and should question rationale of the African market given availability of infrastructure, political stability, etc.

Difficulty: Hard **Quant, Graph Heavy** **Industry: Fin. Services, Outsourcing**

Type: Cost Reduction

Discussion Questions & Solution

Questions for the candidate

- Question 1: What elements would you consider when evaluating the 3 countries?

Solution Guide

Financial

- All of the following should be focused on isolating the change from current to future to isolate the costs savings
- Operating costs (Labor, Rent, Utilities, Transport for employees, Overhead)
- Investment Cost (Important that this can be recovered over a reasonable amount of time)

Other Considerations

- Firm
 - Alignment with firm strategy
 - Experience in Africa
 - Opportunity Cost
 - Risk to customers (quality)
- Market (Africa)
 - Availability of Labor (English speaking, Banking knowledge)
 - Political stability
 - Availability of Infrastructure (Internet, Electricity, other basic needs)
- Competitors – have they already done this?

Discussion Questions & Solution

Questions for the candidate

- Question 2 (Focused on Costs): What are some typical costs associated with running a call center? Which of these would be lower in Africa?

Solution Guide

- Successful candidate will leverage key operational costs from structure
- Key insight is to identify (push candidate to drive towards Exhibit 1) that labor will be the key savings from the initiative (all other costs will remain the same)
- **Share with candidate that existing cost is \$60M per year**
- Candidate should move towards recognizing the number of calls that are made by the call center per day and use that to determine the number of employees required and use the new labor costs to isolate the differences
- In order to be just as effective, the same number of calls, must be made by each call center (i.e. Utilization)
- **Strong candidate will recognize that new call centers will not be effective but interviewer should push candidate in that direction and share Exhibit B**
- Candidate should work through solution to determine that financially, Country B is the best option. A strong candidate will note this immediately, as the effectiveness of A and B are the same, but the cost is lower by \$2/hr
- Strong candidate will recognize that Utilization should increase over time (ignore during calculations)
- Bonus: Security may be an added cost given political environment in some countries – acknowledge but inform candidate it is included
- Bonus: Cost of infrastructure may be higher – acknowledge but inform candidate it is included

Discussion Questions & Solution

Questions for the candidate

- Question 3: If the investment in each country is \$36M, what is the payback period for the country the candidate selected?

Solution Guide

- This question will test basic finance concepts of payback period
- Strong candidate will mention discount rate (candidate can ignore it in calculations)
- Are there ways the Call Center can generate revenue?
 - Candidate should brainstorm possibilities for revenue generation from the call center:
 - Cafeteria that sells food to employees
 - Day Care facilities for working parents
 - Alternate use of facilities as a training center
 - Advertising revenue from posters, etc.
 - Internet café (leverage infrastructure)
 - On-site bank/ATM
 - Vending machines
 - Gym

Math Solution

Call Demand	Current	Country A	Country B	Country C
Employees	400	600	600	1200
Avg. Call Duration (min/call)	4	4	4	4
Total Working time (hrs)	8	8	8	8
Utilization	75%	50%	50%	25%
Total Calls	36000	36000	36000	36000
Employee Cost	\$ 24,000,000	\$ 14,400,000	\$ 11,520,000	\$ 17,280,000
Difference		\$ (9,600,000)	\$ (12,480,000)	\$ (6,720,000)
Cost/employee	\$ 60,000	\$ 24,000	\$ 19,200	\$ 14,400
Employee Cost/hour	\$25	\$10	\$8	\$6

Investment Cost	\$ 36,000,000
Savings per year	\$ 12,480,000
Payback period	2.88

Conclusion

Recommendation

- Client should move it's call center to Country B as it is the most cost effective based on employee rate and productivity
- Risks:
 - Political Stability in the region
 - Quality of Service – will it remain the same?
 - No experience in Country B
 - Long term ability to ensure low hourly rates

Next Steps

- Work with local governments to gain support for investment
- Consider pilot/phased approach to address service quality
- Work towards identifying other revenue opportunities and cost savings initiatives to offset any rise in labor costs
- Implement measures to increase utilization of Country B employees

Exhibit 1

Typical Cost Structure for Call Center

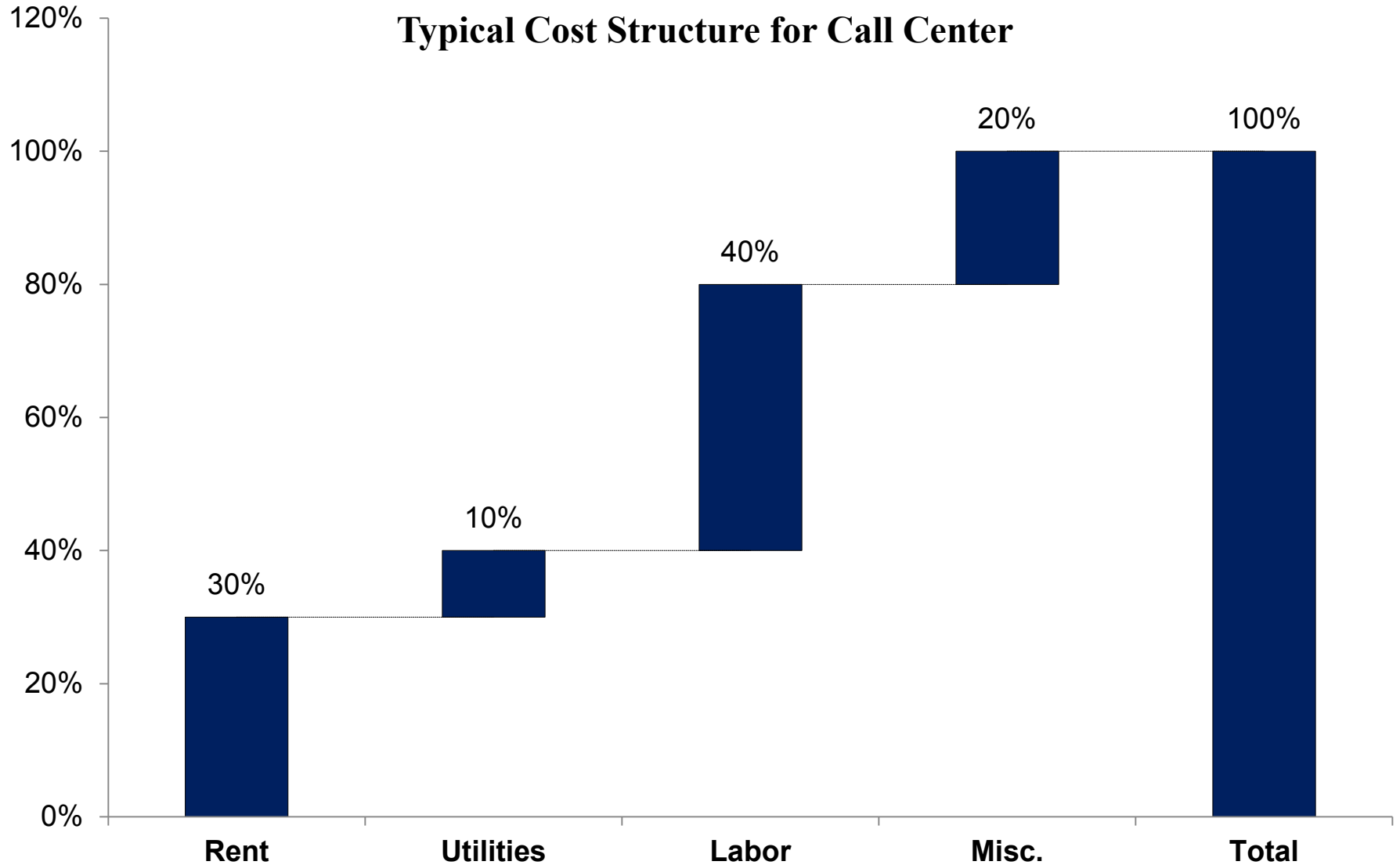


Exhibit 2

Call Demand	Current	Country A	Country B	Country C
Utilization Rate	75%	50%	50%	25%
Labor Rate	\$25/hr	\$10/hr	\$8/hr	\$6/hr
Avg. Call Duration (min/call)	4	4	4	4
Total Working time (hrs)	8	8	8	8

Current call center has 400 employees

Assume 300 days/yr. in calculating annual cost of labor

Burger Chain (Private Equity)

Ross Original

Problem statement narrative

- Our client is a private equity firm. They have come to us for guidance regarding a possible acquisition of a large fast food burger chain. Should they make the acquisition? What should they think about?
- *3-year investment period.

Guidance for interviewer and information provided upon request

- Fast food chain is 3rd largest in U.S. (no non-US operations)
- Operate 5,000 locations; corporate owns all the real estate associated with locations
- All locations are franchised
- Private equity client's portfolio consists of multiple residential/commercial real estate holdings as well as a portfolio of gas stations across the U.S.
- PE firm requires a 15% return on any investments it makes

Difficulty: Medium

Quant Heavy

Industry: PE, Hospitality

Type: Acquisition

Possible Structures

Guidance for interviewer

- A great structure will include aspects regarding the acquisition price, a historical/projected financial snapshot of the fast food company, and an industry piece.
- An excellent structure will also look to explore auxiliary pieces such as franchise agreement implications and exploration into the fit within current PE company's holdings could be explored as well.
- Once structure has been presented, ask candidate where they would like to start:
- Answers to possible questions post structure:
- Industry – Outlook is favorable, not many new players are entering the mature market
- Historical financial results of Burger Chain have been fine.
- Franchise agreement structure is great to explore but not relevant for this case.

Solution for brainstorming financial aspects

Solution Guide (require interviewee to brainstorm before providing answers)

(Pose this question to interviewee)

- Ok, now let's brainstorm some of the financial aspects associated with this company's structure, what components go into the revenues and costs. How would our PE client receive cash flow?

Burger Chain's financial structure

- Revenue (require candidate to brainstorm revenue pieces stemming from franchised locations prior to giving the information)
 - Franchised locations provide the corporate office with the following revenue pieces :
 - Rent (as a % of sales) in this case 15%
 - Royalties (as a % of sales) in this case 3%
 - National advertising quota (as a % of sales) in this case 2%
 - Initial franchise fee (\$100K) for new stores
- Costs
 - Franchised locations – all operating costs are assumed by the franchisee , corporate office – rent for stores and headquarters, SG&A, labor, insurance, legal, etc.

Suggested Solution and Structure

Solution Guide

Year	1	2	3
Average Store Sales (give to interviewee)	\$ 500,000.00	\$ 500,000.00	\$ 500,000.00
New Franchises (give to interviewee)	100	100	100
<i>Assume all new locations generate the average sales figure in the year they are brought on</i>			
Revenue			
Rent	\$ 75,000.00	\$ 75,000.00	\$ 75,000.00
Royalties	\$ 15,000.00	\$ 15,000.00	\$ 15,000.00
Advertising	\$ 10,000.00	\$ 10,000.00	\$ 10,000.00
Average per store revenue	\$ 100,000.00	\$ 100,000.00	\$ 100,000.00
Locations	5,100	5,200	5,300
Total (3 revenue pieces * # of stores)	\$ 510,000,000.00	\$ 520,000,000.00	\$ 530,000,000.00
Franchise fees (\$100K * new stores in that yr)	\$ 10,000,000.00	\$ 10,000,000.00	\$ 10,000,000.00
Total	\$ 520,000,000.00	\$ 530,000,000.00	\$ 540,000,000.00
Total Costs (give to interviewee)	\$ 507,000,000.00	\$ 517,000,000.00	\$ 526,000,000.00
Cash flow to PE firm	\$ 13,000,000.00	\$ 13,000,000.00	\$ 14,000,000.00

Suggested Solution

Solution Guide

(Pose this question to the interviewee)

- Now assume the PE firm knows it can sell the business for \$600M at the start of year 4. Assuming a zero percent discount rate, does the return on the investment meet the threshold set by our client?

Sell to Buyer at the start of Year 4	\$620,000,000
Cash flow from 3 years of operation	\$40,000,000
Total Return	\$660,000,000
Initial Investment	\$550,000,000
Return on Investment	20%

Yes $20\% > 15\%$ required by client. It is not necessary for the interviewee to calculate the exact return, just important that they realize it exceeds the threshold.

Conclusion

Recommendation

Acquire Burger Chain

- Exceeds return requirement
- Doesn't quite fit with current portfolio, however, the real estate aspects of the deal are somewhat of a fit
- Risk associated with exit in year 4 (it's not guaranteed we can sell), however the annual cash flow figures are good

Next Steps

- Negotiate with seller for better price (lowering the initial investment possibly)
- Attempt to keep high level managers in place post-acquisition (contracts, etc.)
- Identify synergies with portfolio companies

Fitness Start Up – Kiosks

Ross Original

Problem statement narrative

- The client is the CEO of a healthcare / fitness startup company that has built a new kiosk related to weight management, whereby there are user accounts and people can track their weights, set goals etc. while on the kiosk, as well as access the information online. They have already Built this product.
- The Client is curious about whether and how to enter the market and has hired you to advise him on this.

Guidance for interviewer and information provided upon request

- Clients objective is be profitable in year 1

Difficulty: Medium

Quant, Graph Heavy

Industry: Fitness-Lifestyle

Type: Market Entry

Possible Structures

Solution Guide

Sample Framework:

EXTERNAL

Markets & Market Sizes

1. Health Clubs - 35k in US.
2. Corporate Wellness Programs

Competitors

Mechanical Beam Scales, Consumer Internet Scales, and Electronic Scales

Customers

End-users, and market people

INTERNAL

Product

Price

Placement/Distribution - See EX 1

Promotion

Branding Stuff - how is that relevant (maybe as a next step)

Brainstorming financial aspects

Solution Guide (require interviewee to brainstorm before providing answers)

	<u>US Health Clubs</u>	<u>Hospitals</u>	<u>Corp Wellness Programs</u>
# locations	35000	6000	unknown
# units/location	3	100	unkown
Price/Unit	1000	500	
Derived TAM	\$105,000,000	\$300,000,000	
<i>Ask Candidate what they think is the value proposition in each location before giving the answer</i>			
	Member Retention	Unclear maybe nurse efficiency	Unclear maybe healthier employees or lower insurance cost
	(industry has 40% attrition)		
	This is a MAJOR pain point	This is a WEAK value prop	This is a WEAK value prop
Ideally only provide # locations to interviewee and they should ask for the rest. Maybe also give them price differential, as it should prompt them to ask re units/location			

* **TAM: Total annual market**

Brainstorming

Solution Guide

Business Model

Have the candidate brainstorm possibilities:

- 1-time upfront fee
- 2-subscription service
- 3 - Other?

Pricing

Have the candidate explore options for pricing on open-ended basis

If asked, COGS = 500

Other Health Club comparable products are priced \$1500-\$3000

Purpose here is to test brainstorming, you can later provide them w data in Ex3 that answers these things

Calculations

Solution Guide

Exhibit 1:

<u>Type (4)</u>	<u>Tradeshows</u>	<u>Online Leads / SEO</u>	<u>Magazine Ad</u>
Cost	\$5,000	\$20,000	\$5,000
# Eyeballs	1,000	15,000	10,000
% Eyeballs who buy	1.00%	0.50%	0.01%
Price/Unit	\$1,000	\$1,000	\$1,000
Hits in 12 month period	3	1	3

Calculations

Rev/Year	\$30,000	\$75,000	\$3,000
Cost/Year/Distribution	\$15,000	\$20,000	\$15,000
COGS	\$15,000	\$37,500	\$1,500
Net Income	\$0	\$17,500	-\$13,500
NI Without COGS	\$15,000	\$55,000	(\$12,000)

Conclusion

Recommendation

- The client should enter market via healthcare market
- Online Leads / SEO only
- Net Income or Profit of \$17,500 whereas Tradeshow will Net Flat and mag ads will be negative

Risks / Next Steps

- Website crash. Technical difficulties
- exposure to competitor
- don't gain brand recognition from other activities that could serve as loss leader
- Next Steps - move into online lead/seo marketing efforts

Barlly Inc. Cosmetics

Ross Original

Problem statement narrative

Your client is Barlly Inc., a cosmetic company based in the US. Barlly has business both in the US and globally. Currently, the client is facing a high level of inventory in the US and is hiring McKinsey to help solve the problem.

Guidance for interviewer and information provided upon request

- The client has had the inventory problem for quite a while, so it is a structure problem other than an one-off problem.
- The client sources raw materials from all over the world.
- The client has one manufacture site near Chicago and 4 distribution centers across the US.
- The client maintains its inventory at DCs, i.e. when products are manufactured, they are directly sent to DCs.
- The client sells to all kinds of retailers across the US.
- Sales is seasonal with peaks before major holidays.

Difficulty: Hard

Conceptual

Industry: CPG

Type: Operations

Additional Questions to Steer Discussion

Questions for the candidate

- How can the client reduce its inventory in the short-term?

Solution Guide

- Reduce production. (This is important)
- Increase sales. Some possible ways:
- Reduce price
- Use outlet
- Volume discount to buyers
- Marketing promotion
- Dump to overseas market

Additional Questions to Steer Discussion

Questions for the candidate

What are possible reasons for the high inventory level and how can the client improve in the long round?

This is a brainstorming case. Every step in the value chain can go wrong, so the interviewer can discuss any step in the value chain according to interviewee's reaction and probe deeper.

The purpose of the case is to see:

- How the interviewee comes up with a structure that can cover the whole value chain.
- How the interviewee use hypothesis driven method to discuss possible reasons for the problem.
- How the interviewee ask for information and use that information to prove or disprove the hypothesis.

Suggested Solution and Structure

Solution Guide

- The interviewee should come up with a value chain that describe the end-to-end supply chain of the client and analyze step by step through the value chain.

Forecast

- The client has an average forecast error of +/- 25%, which means improving forecast can improve future inventory level.
- The closer to the sales date, the more accurate the forecast is. Therefore, the client should try to use the latest possible forecast number for supply chain. However, the lead time for some of the raw material is too long. Suggest to negotiate with suppliers to shorten lead time or build stock for such raw materials.

Suggested Solution and Structure

Solution Guide

Procurement

- Some raw materials have very high minimum order quantity (MOQ) and large purchase for such materials built up inventory. Negotiate with suppliers to lower MOQ.
- Some suppliers often miss delivery time and production window. Production is delayed and other raw materials are sitting in the inventory and waiting for the missing ingredients. Improve supplier compliance or switch suppliers.
- Procurement department is only evaluated by service level, so they tend to build up inventory to ensure material availability. Introduce inventory level into the evaluation system.

Manufacturing

- There is mechanical constrain that a large quantity has to be made in a run (large batch size). Invest in new equipment and R&D to reduce batch size.

Suggested Solution and Structure

Solution Guide

Distribution

- Safety stocks are maintained at every DC. Centralize inventory at manufacturing site can reduce total safety stock by half. (square root of number of DC).
- The replenish cycle is currently a week. Increase replenish frequency could reduce cycle stock.

System Overall

- The client puts high emphasize on service level, so every step in the value chain increases its “safety stock” to ensure high service level. Inventories then add up. Suggest to only maintain safety stock at finished goods level and eliminate “safety stocks” at all the other steps.
- Bring inventory management mind set to clients’ employees and provide proper tools for employees to manage inventory level.

Conclusion

Recommendation and Next Steps

- The client should reduce production and dump inventory in the short round.
- Systematically improve supply chain in the long round. Recommended solutions depend on what were covered during the case process.
- Employee mind set change.

Booz Round 1: PE Services Firm

Problem statement narrative

- Your client USCo is a firm that provides services to private equity firms in the U.S. It recently acquired EUCo, a similar company in Europe.
- How can they realize synergies both quantitatively and qualitatively over the next 1-2 years?

Guidance for interviewer and information provided upon request

- How large are the US and EU businesses? –
 - 5M and 4.5M respectively
- What was the reason behind the acquisition?
 - Entry into European markets

Difficulty: Medium

Industry: PE

Type: Merger Effectiveness

Additional Questions to Steer Discussion

Questions for the candidate

- Direct candidate away from revenue increases and other non-cost synergies – the focus should be cost savings.
- Initial findings indicate that the bulk of the cost savings come from IT. Cost structure for the two companies are in the exhibit.
- Focus on licensing and labor (FTE) costs: the main differences between USCo and EUCo is that USCo develops its own software (hence the high labor cost and no licensing fee) whereas EUCo purchases its software from software firms (hence lower labor cost but substantial licensing fees).
- Indicate that the two main approaches being considered are to adopt either the USCo approach or the EUCo approach. Get candidate to determine which is quantitatively preferable.
- If USCo adopts the EU approach, there will be an additional 1B in licensing fees while FTE costs would drop from 1.8B to 0.5B.
- If EUCo adopts the US approach, there will be an additional 0.4B in FTW costs while licensing costs would drop to 0.

Suggested Solution and Structure

Solution Guide

Option A: Adopt USCo approach in EU

Original licensing + FTE costs = $1.5B + 0.7B = 2.2B$

New costs = $0B + 0.7B + 0.4B = 1.1B$

Cost savings using A: $2.2B - 1.1B = 1.1B$

Option B: Adopt EUCo approach in US

Original licensing + FTE costs = $0B + 1.8B = 1.8B$

New costs = $1B + 0.5B = 1.5B$

Cost savings using A: $1.8B - 1.5B = 0.3B$

Quantitatively, Option A is preferred. However, qualitative issues to note:

- Possible long-term software supplier contracts in EU
- Potential additional costs to localize US software in EU
- Need to train personnel in EU; potentially sending US personnel to train local staff
- Other factors to reduce the other IT cost buckets are good (e.g. adopting cloud computing to lower HW costs)

Conclusion

Recommendation

Go with Option A and adopt the US model in EU due to the quantitative costs savings of 1.1B vs. 0.3B for Option B

Next Steps

Determine additional migration and localization costs
Verify if EU customers can use the same software as US customers
Identify potential regulatory concerns using US software in EU

Costs Breakdown

IT Costs	USCo	EUCo
Licensing Fees	0	1.5B
Hardware	300M	800M
FTE (Labor)	1.8B	0.7B
Others	600M	400M

PD Gas Buyout

Based on BCG & McKinsey, Round 2

Problem statement narrative

- Your client is PD Inc., a large US based grocery supermarket chain. PD Inc. also runs 999 gas stations next to its retail stores.
- Last week, PD Inc. was approached by a large US based oil and gas distributor which offered to buy out the entire portfolio of 999 gas stations from your PD Inc.
- Your PD Inc. immediately reached out to you and has sought your advice on whether to sell these gas stations or not and what factors to consider when making this decision.

Guidance for interviewer and information provided upon request

This is a interviewee driven case in the first half and an interviewer driven case in the second half. A few answers to general questions:

- PD Inc. has not been offered a specific price by the buyer. Interviewee can be told to consider price as part of his recommendations though. The key is to first decide whether the PD Inc. should sell or not.
- PD Inc. cannot choose to sell a part of the portfolio of gas stations. It will either sell the entire portfolio of 999 gas stations or nothing at all.
- No information on competition is available.
- No information on the buyer's scale, geographic presence or reason to buy is available
- PD Inc.'s stores are spread across the United States. All stores have a gas station next to them.

Difficulty: Hard

Industry: Retail

Type: Divestiture

Information for candidate to be provided upon request

- A good candidate will identify that initially, PD Inc. needs to value its gas station business based on cash-flows and also identify the synergies with the retail stores. Information to be provided upon request:

	Revenue (In \$ MM)
Revenue for each gas station each year	60
Cost of fuel sold at gas stations (COGS)	59.98
Cost of licensing fees paid to distributor	3.02
Cost of rent of land and equipment	2
Cost of labor, utilities, insurance and miscellaneous	1
Net Profit	-6

- Really good candidates identify that the Gross Margin is extremely low either because it's a very low margin business or because PD Inc. is discounting prices on purpose to attract customers. In this case, PD Inc. is using its fuel stations as loss leaders
- Candidate should ideally then enquire about retail revenue and synergies.

Answers to additional questions

- PD Inc. has been making losses in the gas station business since the past 3-4 years
- There is no information on the growth of the fuel and the retail business for PD Inc.
- The gas industry as a whole has a similar cost structure
- The gas station has no additional revenue streams (carwash/repair/convenience store)
- PD Inc. has no other business verticals other than fuel and grocery retail

Solution Guide: Revenue side

- For revenue on the retail (grocery) side, upon request, candidate should be told that fuel station revenue accounts for 20% of overall revenue. Hence, retail revenue will be \$240 million per store.
- Upon request, candidate should also be told that we are very proud of the way we manage our suppliers and have fairly high profits margins relative to the retail (grocery) industry. The margins on the retail (grocery) side are 16.66%. Candidates intuitively good with numbers will identify this as the fraction $\frac{1}{6}$. if they don't, tell the candidate to consider it as $\frac{1}{6}$.
- The consolidated revenue and cost figures for PD Inc.'s business are given below:

	Revenue (In \$ Mn)	Cost (In \$ Mn)	Profit (In \$ Mn)
Gas Stations	60	66	-6
Retail Stores	240	200	40
Total	300	266	34
After Sale	204	170	34

Solution Guide - Synergies

- Potential synergies are additional walk-ins to retail stores, joint loyalty program, supply chain synergies (cheaper fuel for PD Inc.'s trucks, same trucks used to deliver goods etc.)
- Additional information on synergies:
 - 15% of Inc.'snc's customers on the retail side come to buy groceries only because they came to the gas station to fill up gas
 - 40% of PD Inc.'s customers on the retail side come to buy groceries but also end up buying gas at the gas stations later (i.e., they don't really care if there was no gas station next door)

Candidate should realize that the 15% of the customers who came to buy gas first are the one which account for synergies on the retail side directly attributable to the gas station

	Revenue (In \$ M)	Cost (In \$ M)	Profit (In \$ M)
15% synergy impact	36 (15% of 240 Mn)	30	6 (16.66% of revenue)

- **Candidate should now realize that the losses on the fuel business are getting covered by the synergies from the fuel business on the retail side. Hence, it is a wash.**

Suggested Solution and Structure

Solution Guide

- Now is when you turn the case into a discussion and ask the candidate to evaluate the positive and negatives for PD Inc. if it chose to sell the business. It is important to keep questioning the candidate's assumptions.

Positives from Sale	Negatives from Sale
Increased cash flow due to cash received from sale	Drop in economies of scale as PD Inc. no longer buys \$30 B worth of goods. This also results in excess warehouse, transportation, store capacity hence downsizing costs
Lesser working capital (reduced by \$5.5 B --- Annual cost of \$66Bn divided by 12 months = approx. \$5.5 B in working capital)	These 15% customers will go to competition which will get economies of scale hence lower prices and this may result in further erosion of our customers
Increased focus on existing business	Stock market may react negatively to drop in revenue
Leaner operation with higher margins (refer to next slide for details)	May impact inventory turnover etc.

- It is important to keep asking candidate for more positives and negatives and question the candidate. “Are there really any negatives from the sale since there seems to be no impact on profits and for a business, profits are the best metric to gauge impact?”, “Why will the 15% customers not come to the stores any more?” Etc.

Solution Guide – Loss of Revenue after transaction

- Most candidates will assume that PD Inc will lose the 15% of the customers once it sells the business but it is important for them to identify the exact reason why PD Inc will lose these customers.
- Question the candidate's assumption by stating that the gas station is still next door so why will the customers stop coming.
- The correct reason is that PD Inc is currently selling the fuel at discounted prices to get customers to come to the fuel station and then buy groceries but the buyer has no incentive for doing so and is likely to raise prices to market levels and hence the customers will stop coming to the gas station.
- You can further question the candidate by showing him the resulting expected impact on margins after selling the business. The actual impact will differ as this margin does not include downsizing and other costs. These figures are given to candidate only to see if he/she can realize that the actual figures may be very different because of the negatives listed in previous slide. The margins as %age of revenue are given below:

	Revenue	Cost	Profit
Pre-sale	100%	88.66%	11.33%
Post-sale	100%	83.33%	16.66%

Suggested Solution and Structure

Recommendation

An ideal recommendation is to advise PD Inc to sell the business but contractually obligate the buyer to three conditions:

1. The buyer will keep all fuel stations open/seek approval before closing stations
2. The buyer will not open any competing establishment (convenience stores) at the fuel stations
3. The buyer will keep the prices discounted by allowing PD Inc to subsidize the prices. (the key is to realize that PD Inc does not have to discount so heavily since fuel is highly price elastic. Even if the buyer sells fuel at prices 5% below competition (assuming competition sells to entirely breakeven, i.e., at \$66 Bn, buyer will sell it at ~ 63 Bn) the customers will still come. PD Inc. can reimburse ~ \$ 3 B to buyer and hence still end up making a net profit of ~ 3 B on the retail side along with getting benefits from all the positives of selling the business.

Next Steps

1. Conduct a market survey to test price sensitivity of customer
2. Discuss the stipulated conditions with buyer. The 40% of customers that end up buying fuel can be used as a leverage
3. Discuss potential for co-branded loyalty card to further increase customer overlap

Ocular Implant

BCG, Round 2

Problem statement narrative

- Our client is a small medical startup company that has invented a new ocular product which it believes will revolutionize the vision for people suffering from degenerative eye disease.
- The product has been patented however is yet to undergo clinical trials.
- The final product would need the patients to undergo a surgery and decrease the number of visits to the doctor. The first two stages of clinical trial is expected to cost our clients \$14M.
- Our client is trying to decide if they should try and sell this technology to a major player or should they raise capital and go to market with the product.

Guidance for interviewer

- Offer following information if candidate asks for it –
- There are currently no buyers for this product although some companies have expressed interest in investing in the technology for future rights

Difficulty: Hard

Industry: Healthcare

Type: New Business Development

Suggested Solution and Structure

Structure

A good candidate will highlight the following points in the structure

1. Trying to differentiate the current treatment from this new ocular implant and the value proposition for the clients
2. Realizing that clinical trials might take years and there is not guarantee for success and eventual FDA approval
3. The tort implications if they decide to go to market individually and also the amount capital that will be necessary to go to market with this product
4. Possibly get funding from a major player to undergo clinical trials since major bio-med companies already spend billions on clinical trial and would not mind spending a few million to see the efficacy of the device
5. Will insurance cover this device

Comparison of current method with new ocular implant

Today patients visit the doctors every quarter; each visit costs them \$350. The disease afflicts about 9% of the US population.

1. Average age when this disease sets in is 30
2. This is a degenerative disease and patients usually have to carry on the treatment for their lifetime
3. Assume an average life expectancy of 60 years
4. The new ocular device implant surgery would be a onetime cost of \$37,250
5. After the surgery the patient is required to visit the doctor annually where the doctor checks and replaces a fluid which costs the patient \$75 per visit

Average cost for traditional method = $60 - 30$ (average years of disease) * 4 (number of visits) * 350 (cost per visit) = \$42,000

Average cost for new method = $\$37,250 + (60 - 30) * 1$ (number of visits) * $75 = \$39,500$

Ask the candidate for his opinion on the cost difference

The cost is lower but not by a very high amount. However, revenues available at the present time is worth more than the same amount in the future due to its potential earning capacity

It is important to determine the patients' perspective as well; although there are fewer visits, does it imply a higher satisfaction? (The client's research indicates that even if the costs were higher patients would prefer the ocular implant for not only does it offer lower visits but also much clearer vision than the traditional method)

Conclusion: What should the client do?

1. The company should approach major bio-med companies for investment to get the clinical trial and FDA approval underway in return for exclusivity.
2. The company should explore other options of using the ocular implant, e.g. can it be used to deliver another drug or treat another form of disease.
3. It will be difficult for the company to go to market by itself because it would necessitate a huge infrastructure and hence they should value their product and sell their patent post clinical trials for exit

Shipping Company

McKinsey, Round 2

Problem statement narrative

- Our client is a major shipping company and owns 200 ships and leases an additional 150 ships. Recently they have noticed a steep decline in their profitability and would like to find out how to remedy the situation.
- Our shipping company transports all kinds of goods except oil and other liquids. We also do only port to port transport which does not include ground transportation from the port to/from client location

Guidance for interviewer

- The objective here is to have the interviewee walk through the major reason for lower profitability, primarily declining revenues and increasing costs.
- A good structure would cover the following points (see next page)

Difficulty: Medium

Industry: Transportation

Type: Profitability

Suggested Solution and Structure

Reasons for decline in Revenues

1. Decrease in global trade owing to the slowdown in Europe and China
2. Increased competition
3. Decrease in utilization (load factor) of ships
4. Emergence of alternative transport mechanisms
5. Types of service offered vis-à-vis product mix (e.g. shipping time vs. price)
6. Brand image problems causing declining revenues
7. Other problems associated with reputation e.g. recent accidents
8. Security and reliability of delivery
9. Competitors offering ground transportation in addition to port to port transport

Reasons for increase in Costs

1. Increase in maintenance for ships
2. Increase in insurance costs
3. Labor and material costs
4. Fuel costs
5. Costs associated with route (e.g. Piracy resulting in longer hauls or weather related changes causing longer time to ship)
6. Environmental costs
7. Costs resulting from underutilization of ships
8. Increase in leasing costs
9. Duration and contract terms for leased ships
10. Holding costs related to holding items in port
11. Docking and transit costs paid to governments

What would have been the impact on profits if fuel prices did not rise?

Our client conducted an in depth analysis and discovered that it's cost for the current fiscal year were \$1B. The revenue has remained steady but the load factor of ships has decreased. 35% of this cost directly resulted from payment made for fuel. The price of fuel has appreciated 300% since 2009. Assuming that the fuel price did not appreciate what would be the net impact on our bottom line.

Fuel cost in 2012 = \$350M

The fuel appreciated by 300% therefore fuel cost in 2009 = \$87.5 M

If the price today was \$87.5 M it would have meant an increase in profit of $(350-87.5) = \$262.5$ M

Brainstorm: Given our discussion so far, how can profits be increased?

1. Given that revenue has remained steady and load factor has decreased implies we are making more runs, we should identify ways to increase our load factor and decrease the number of runs
2. This would mean holding goods at the port for a longer duration
3. It is interesting that the fuels costs have increased by 300% whereas the oil prices in the market have not seen a similar increase indicates the price must be a result of the refining cost. It will be important to investigate the reason for increase in refining costs.
4. We could renegotiate our lease contracts and possibly terminate some of these
5. We could hedge fuel (make sure the interviewee understands how and what they will be hedging – if the refining cost increase is not permanent the hedging would be detrimental)
6. Explore replacing engines with ones that use electric motor's
7. Optimize routes to take advantage of ocean currents thus saving on fuel and decreasing transit time
8. Ensure optimum vessel size and optimize route so that there is commonality between refueling stops and port calls for transfer of goods.

Dog Daycare

Ross Original

Problem statement narrative

- Your client is a dog daycare based in the northeast U.S. She has recently been approached by a pet retail chain requesting to buy her business.
- Should she sell it? If so, how much should she ask for the business? If not, what should she do to grow the business herself.

Guidance for interviewer and information provided upon request

- Your client currently has only one store
- Their services include daycare and grooming
- The retail chain is not capital constrained
- Daycare capacity is 80 dogs a day. Current utilization is 60% on weekdays, 40% on weekends. They charge \$40/day for daycare
- Grooming capacity is 10 dogs a day. Current utilization is 70% on weekdays, 10% on weekends. They charge \$60 for grooming
- Assume 250 weekdays/year and 100 weekend days/year
- Use a 10% discount rate

Difficulty: Medium

Industry: Pet Care

Type: Strategy, Valuation

Additional Questions to Steer Discussion

Questions for the candidate

- 1) How would you value this business?
 - Look at current free cash flow, apply a discount rate and assume it is a perpetuity
- 2) Assuming you decided not to sell the business, how else could you grow your business?
 - Add new products/services such as: training, boarding, retail, walking, vet services
 - Look for opportunities to open new stores/acquire existing stores in the area
 - Seek outside investment to expand

Suggested Solution and Structure

Solution Guide

1) Client's current revenue/costs

- Revenue
 - Daycare Weekdays = 48 dogs/day x \$40/day x 250 days=\$480,000
 - Daycare Weekends = 32 dogs/day x \$40/day x 100 days=\$128,000
 - Grooming Weekdays = 7 dogs/day x \$60/day x 250 days=\$105,000
 - Grooming Weekends = 1 dog/day x \$60/day x 100 days=\$6,000
 - Total=\$719,000
- Costs
 - Labor = 3 employees/day x \$10/hr x 12 hours/day x 350 days/yr=\$126,000
 - Rent = \$12,000/month x 12 months=\$144,000
 - Supplies = \$2,000/week x 50 weeks/yr=\$100,000
 - Insurance = \$30,000/yr
 - Total=\$400,000
- Subtract \$99,000 to account for taxes and other items, resulting in free cash flow of \$220,000

Suggested Solution and Structure

Solution Guide

2) Acquisition Details

- Current free cash flow is \$220,000
- Assume no growth in free cash flow
- Assume the free cash flow represents a perpetuity
- Assume 10% discount rate
- Present value of the perpetuity= $\$220,000/10\%=\2.2M is the present value of the client's business
- Since the client is negotiating with the retail chain, she should ask for a premium of ~20%
- The client should offer to sell her business for \$2.64M

Suggested Solution and Structure

Solution Guide

3) Client grows business organically

- Ask interviewee to brainstorm new products or services that could be added
- Assume new products and services could add \$400,000 of free cash flow/year
- The store's new free cash flow would be \$620,000
- After the new services are added, the value of the perpetuity goes to \$6.2M, far greater than the \$2.64M the client was originally hoping to get for the business

Conclusion

Recommendation

- Try to get the retail chain to pay at the higher valuation, despite not having the products/services in place yet
- If the chain is not willing to pay the higher value, client should implement new products/services on their own
- With additional cash, client could look to acquire new stores on her own

Next Steps

- Negotiate with retail chain
- Create business case for implementing new products/services
- Investigate competing stores in area to see if they are open to acquisition by the client

Seaworthy Floating Hotel

Ross Original

Problem statement narrative

- Your client is a successful entrepreneur based in Manhattan. She is always thinking of creative ways to start new businesses. Her latest idea is to build a floating hotel . (Think of a skyscraper on a barge, and the combined structure is seaworthy.)
- Today is Oct.1, and she must make a decision ASAP – her preferred construction company and barge manufacturer conveniently have openings on Nov.1 for projects. If she misses this window it will be another two years before both companies can simultaneously work together.
- The entrepreneur hires you to assess whether she should spend \$20M to build a ‘floating hotel that is seaworthy’

Guidance for interviewer and information provided upon request

- If the candidate finds the case odd, let him/her know the entrepreneur’s success rate is 100%, and the hotel project is vapid compared to previous endeavours
- Potential clarifying responses (only provide if asked):
 - The hotel can travel on water like a sea vessel
 - Time for construction and validation: 1 year
 - Location for hotel & barge fabrication: New Jersey
 - \$20M is a sunk investment and represents all labour, material and testing costs to manufacture and validate the seaworthy floating hotel
 - The financial objective is 20% ROI in the first year
 - She wants to begin operating the hotel as soon as construction and testing conclude; that is, Nov.1 of the following year
 - Assume no construction delays
 - Assume the client has sufficient funds

Difficulty: Hard Quant, Graph Heavy Industry: Hospitality

Type: New Business Development

Part 1: Background Info and Basic Market Analysis

Market Analysis

- The candidate may want to start with ROI. Guide him/her to the Market Analysis first
- Please ensure the candidate explains why s/he wants information before providing it
- Market Analysis
 - Travel & Tourism Industry: good growth in tourist cities
 - Competitors: there are no other seaworthy floating hotels in the US (east or west coast)
 - Other Competitors and Substitutes: numerous & fragmented – chain hotels & motels, small business operators, bed & breakfast, campers/RVs, weekly apartment rentals, etc.
 - The exact size of the industry is unknown, ditto for market share
- The client is only considering the US market (east coast to be specific)
- Government regulations will be addressed and met during the construction and validation phases
- Once the candidate sufficiently addresses the market and any potential regulations, ask him/her what location(s) the client should consider for the hotel?
(Proceed to the next slide)

Part 2: What Location(s) Should the Client Consider?

Where to Locate?

- The aim here is to test the candidate's business knowledge of the travel & tourism industry; specifically, high and low seasons (in other words, the seasonality effect)*
- The candidate's objective is to identify two locations that take advantage of seasonality (recall, the hotel is seaworthy and can travel to different destinations via the Atlantic ocean)*
- Please ensure the candidate explains why s/he wants certain data before providing the information*

Possible Candidate Questions	The Interviewer's Response
How many locations is the client considering?	Two (no more, no less)
How many trips a year will the hotel make?	Two (no more, no less)
At what time during the year will the trips occur?	The client is looking for your guidance (If required, guide the candidate towards seasonality)
What locations is the client considering?	1st Response) What do you think? 2nd Response) How can the client maximise revenue with different locations? (Here, the candidate should combine locations and seasonality to maximise revenues. Specifically, the locations are <u>Miami</u> and <u>Manhattan</u>)
When is high season in Manhattan?	What do you think? (In general, the candidate should indicate late spring, summer and early fall. The specific response is May 1 to October 31)
When is high season in Miami?	What do you think? (In general, the candidate should indicate late fall, winter and early spring. The specific response is November 1 to April 30)

- Ideal Conclusion / Synthesis from Candidate: The client should locate the hotel in Miami from 11/1 to 4/30 and Manhattan from 5/1 to 10/31. Both periods are high season and will maximise revenues
- Next, guide the candidate to determine if the seaworthy hotel meets the client's financial target*

Part 3A: Will the Hotel Meet the Client's Objective?

Data for Revenues

- Ask the candidate how s/he would assess ROI
- Part A: After a sufficient explanation, start with revenues
- Please ensure the candidate explains why s/he wants information before providing it

Possible Candidate Questions	The Interviewer's Response
How many floors does the hotel have	26
How many floors have rooms	Floors 2 through 26 have rooms (25 floors have rooms)
Is there any room segmentation	Good question! Assume all rooms are the same.
How many rooms per floor	Each floor layout is the same: a five-by-five grid (Candidate should immediately calculate 25 rooms per floor)
What is the room's daily rate	High season: \$200 / room / day (both cities) Low season: \$133 / room / day (both cities) /1
Hotel occupancy rate	High season: 80.00% (both cities) Low season: 53.67% (both cities) /1
What's on the first floor	Restaurant, bar / discotheque, convenience shop
In aggregate, what is the first floor's revenue	High season: \$200,000 / month (both cities) Low season: \$ 97,613 / month (both cities) /1
How many days does the hotel operate	The entire time it is docked in the city
How many days does it take to travel from Manhattan to Miami (and vice versa)	2.5 days in either direction Candidate should recognise the hotel operates 360 days a year, and

/1 Low season numbers are provided to distract the candidate

- Proceed to the next page for the annual revenue calculation

Part 3A: Will the Hotel Meet the Client's Objective?

The Top Line: Calculating Total Annual Revenue

- *Part A: Continued... Annual revenue calculation*
- *Ideally, the candidate will realize the computation only involves high season. Additionally, revenue is identical in Miami & Manhattan and, as such, calculations can be shortened*

Annual Room Revenue	=	Rooms per Floor	X	Floors w/ Rooms	X	Occupancy Rate	X	Rate High Season	X	Revenue Days per Season 2/	X	Seasons per Year
Annual Room Revenue	=	25	X	25	X	0.80	X	\$200	X	180	X	2
Annual Room Revenue	=	\$36,000,000										
First Floor Revenue	=	High Season Rev / Month	X	Months per Year	=	\$200,000	X	12	=	\$2,400,000		
Total Annual Revenue	=	Annual Room Revenue	+	First Floor Revenue	\$38,400,000							

2/ Days per Season = {365 days per yr - (2 trips x 2.5 days per trip) } / 2 seasons per yr = 180 days per season

- *Proceed to the next page for the cost analysis*

Part 3B: Will the Hotel Meet the Client's Objective?

Discussing VC and FC

- Part B: Cost Analysis
- Ask the candidate what cost buckets s/he would consider for the seaworthy floating hotel
- For VC, guide the candidate to external house keeping if it is omitted
- For FC, docking fees and insurance are significant. Guide the candidate to these topics if they are omitted

Variable Costs (VC)	Fixed Costs (FC)
House Keeping	Docking fees (significant)
External House Keeping (Daily exterior inspections)	Insurance (massive!)
Food & drink	Yearly regulatory fees
Etc.	Etc.

- Proceed to the next page for the cost data

Part 3B: Will the Hotel Meet the Client's Objective?

Reviewing the Cost Data

- Part B: Continued... Cost data
- Based on the cost bucket discussion, the candidate comprehends the key drivers. Please provide the following data

	Manhattan	Miami
VC	\$150 / day / rm	\$100 / day / rm
FC	\$3M / season	\$1M / season

Note 1: Seasonality doesn't affect VC or FC

Note 2: Federal regulations mandate that
internal & external house keeping
is required regardless of occupancy

- Ideally, the candidate will understand Notes 1 & 2 without further clarification. If not,
 - Note 1 indicates VC and FC are the same in high and low seasons
 - Note 2 indicates VC impacts all rooms every day (assume house keeping is 100% of VC)
- Proceed to the next page for the cost calculation

Part 3B: Will the Hotel Meet the Client's Objective?

Calculating the Costs

- Part B: Continued... Cost calculations
- The candidate should first calculate each city's total cost, and then the total annual amount

Miami Total Cost	=	Total VC	+	Total FC						
Miami Total Cost	=	Floors with Rooms	X	Rooms per Floor	X	VC per room per day	X	Days per Season	+	Fixed Cost
Miami Total Cost	=	25	X	25	X	\$100	X	180	+	\$1,000,000
Miami Total Cost	=	\$12,250,000								
Manhattan Total Cost	=	Floors with Rooms	X	Rooms per Floor	X	VC per room per day	X	Days per Season	+	Fixed Cost
Manhattan Total Cost	=	25	X	25	X	\$150	X	180	+	\$3,000,000
Manhattan Total Cost	=	\$19,875,000								
Total Annual Costs	=	Manhattan Total Cost	+	Miami Total Cost	=	\$32,125,000				

- Proceed to the next page for ROI analysis

Part 3C: Will the Hotel Meet the Client's Objective?

Calculating ROI

- Part C: ROI
- The ROI is calculated as follows:
 - Investment = \$20,000,000
 - Profit = Total Revenue - Total Cost = \$38,400,000 - \$32,125,000 = \$6,275,000
 - ROI = Profit / Investment x 100% = [\$6.275M / \$20M] x 100% = 31.375%
- Client's Target = 20% ROI within the first year
- Once the ROI is complete, instruct the candidate that the client wants the final report in 60 seconds. Please prepare a one page presentation
- One page presentation entails Recommendation, Risks and Next Steps

Recommendation, Risks & Next Steps

Recommendation

The candidate's recommendation should follow a similar format:

- Client's target = 20% ROI within one year
- Resulting ROI = > 31% in the first year

Based on surpassing the financial objective, the client should proceed with construction on Nov.1 to fabricate a seaworthy floating hotel.

The hotel's unique feature to operate as a sea vessel enables it to capitalise on the high tourism seasons in Miami and Manhattan.

Additionally, the client has adequate funding to finance such a project.

Risks

Examples of risks and mitigating factors:

1) Risk of being the first seaworthy floating hotel and not achieving the anticipated 80% occupancy rate

Promotions and advanced sales to secure early bookings. Since the ROI > objective, the client can trim the top line to achieve the desired occupancy rate while maintaining the financial target

2) Competitors will enter the market due to the strong profits. As a result, docking fees may increase and upset the profit equation

Since the client is first to market and has, at least, a one year lead, she should secure a multi year contract for docking fees in Miami and Manhattan

Next Steps

- 1) Ensure contracts are drawn up for hotel and barge manufacturers
- 2) Scout port locations in Manhattan and Miami
- 3) Begin advertising campaign at least 6 months in advance of the hotel's opening

Big Trucks

McKinsey, Round 2

Problem statement narrative

- Our client is a manufacturer and dealer of heavy industrial trucks. They currently offer three brands of trucks. Each brand offers trucks in the light-, mid- and heavy duty class.
- The client currently sells trucks directly to end users and typically includes a clause to repurchase the trucks after a few years.
- What are some areas our client should be concerned about?

Guidance for interviewer and information provided upon request

- This case is a McKinsey 2nd round interview case, it should be given as a command and control style case
 - Candidate should brainstorm aloud – a good candidate will note their hypothesis for future reference
 - Interviewer should push until candidate identifies repurchase agreement, this is the largest factor
 - It is important to review this case **extensively**
- Additional Information
 - Market is composed of three players (Companies not brands)
 - Players are fairly equal in market share
 - Client's brands are not distinguished and direct competitors

Difficulty: Hard Quant, Graph Heavy Industry: Automotive

Type: Profitability

Suggested Solution and Structure

Solution Guide

- Good brainstorming will highlight the following areas:
 - **Repurchase agreement** – Key Driver and Step 1 after brainstorming
 - **Brand proliferation** – opportunity to streamline to one brand
 - **In-house and procured materials** – area for improvement Step 2 after repurchase exercise
 - **Others** – Allow candidate to identify as many as the wish, interviewer should question unrealistic guesses and then have candidate indicate which is likely the most important factor

Repurchase Agreement

Data

- ☐ Length of service – 3 Years
- ☐ Repurchase Price - \$25,000
- ☐ Trucks to be Repurchased
 - ☐ Year 1 – 20,000
 - ☐ Year 2 – 30,000
 - ☐ Year 3 – 40,000
- ☐ Resale Price
 - Year 1 – \$24,000
 - Year 2 – \$22,000
 - Year 2 – \$20,000

Suggested Solution

Repurchase Agreement

- Candidate should note that repurchase agreement creates liability for company
 - Follow up and have candidate size the liability
 - Year 1 – 20K trucks X \$1K loss/truck - **\$20M**
 - Year 2 – 30K trucks X \$3K loss/truck - **\$90M**
 - Year 3 – 40K trucks X \$5k loss/truck - **\$200M**
 - Total Liability of **\$310M**
- Candidate should suggest the following potential solutions
 - Accelerate repurchases to reduced liability (should note this could cannibalize new sales with increases used trucks on the market)
 - If candidate assumes **50% of years 2 and 3 are repurchased a year earlier**
 - Total Savings 35K trucks X \$2K loss reduction/truck - **\$70M** liability reduction – Other feasible acceleration plans are acceptable
 - Increase resale price through refurbishment (only if net positive)

In-house vs. Procured Parts

- Sale Price of Truck - \$50,000
- Total Cost of Truck - \$40,000

Percentage of Parts			
	Client	Competitor A	Competitor B
In-House	40%	50%	25%
Purchased	60%	50%	75%

Relative Costs per Part			
	Client	Competitor A	Competitor B
In-House	100%	80%	105%
Purchased	120%	140%	80%

Suggested Solution

Part Procurement

- Ask candidate to calculate costs of in-house and procured parts
 - In-house - $\$40K \times 40\%$ - **\$16K**
 - Procured - $\$40K \times 60\%$ - **\$24K**
- Have candidate hypothesis why competitors have better costs structures and how client could achieve these:
 - Economies of Scale
 - Attempting to move to price leadership of one would hurt the other
 - Opportunity to increase internal production and sell (Purchase Supplier)
 - Have candidate calculate new price if they were able to match price leader for both in-house and procured parts
 - In-House – Savings of $20\% \times \$16K$ – **\$3.2K**
 - Procured – Savings of $33\% \times \$24K$ - **\$8K** (40% reduction of 120% total is 1/3 savings)
 - **Total - \$11.2K = 28% - Huge opportunity, but is it feasible?**

Suggested Solution

Part Procurement – Implied Costs

Candidates will not be required to calculate competitor costs

	Implied Total Costs		
	Client	Competitor A	Competitor B
In-House	\$16K	\$16K	\$10.5K
Purchased	\$24K	\$23.3K	\$20K
Total	\$40K	\$39.3K	\$30.5K

Conclusion

Recommendation

Client should look to accelerate repurchases to reduce liability. Additionally, client should investigate changing their part procurement strategy. There may be opportunities to reduce costs.

Next Steps

Better understand market elasticity for used/new trucks as these strategies will cannibalize sales from one another.

Interviewer Assessment

It is important for the candidate to note that these strategies could negative impact one another. More readily available used trucks could reduce demand for new trucks, while cheaper new trucks(if margin isn't increased) will make used trucks less appealing and could increase the repurchase liability.

Delson Hard Drive

BCG, Round 2

Problem statement narrative

Your client, Delson, is a \$20B hard drive company that manufactures and sells hard drives to OEMs (e.g., HP). OEMs sell computers and data systems to consumers and all kinds of businesses. Hard drive industry's profitability has been cyclical, and recently seen a decline due to increasing competition and slow sales in computers. Therefore, the CFO of Delson decided to acquire a start-up company, Small Storage Systems (SSS). SSS purchases hard drives and then manufactures and sells data storage equipment for large companies to store and manage data efficiently. Software solutions are included with the equipment.

Now, the CFO meets the BCG Partner at a dinner and asks for advice: what issues should we be concerned about the newly acquired start-up SSS? What should we do?

Guidance for interviewer

- The case presents a very open-ended prompt to test candidate's ability of dealing with ambiguity and focusing on the most important issue (CEO approach).
- Keep pressing the candidate until she/he realizes that how to get customers for SSS is the first step to the success of such a new business.
- Candidate should understand that Delson currently has no sales force to reach large companies, but OEMs have the sales force. Therefore, client can consider building own sales force or leveraging OEMs as a channel. When candidates comes to this point, give Exhibit 1.

Info to be given upon request:

- They are few companies in the OEM space, but many hard drive manufacturers.
- Hard drive industry has a very high fixed cost. Major players have very thin profit margin and need to maximize utilization in order to stay profitable.

Difficulty: Hard

Industry: Computer Hardware

Type: Acquisition effectiveness

Exhibit 1: SSS' Profit Margin via own sales force V.S. via OEM channel

	OWN *	OEM *
List price	100	100
Price to OEM	----	75
COGS	-60	-60
Sales	-15	-5
G&A	-10	-3
Profit Margin	15	7

* All the figures are in %

Suggested Solution

Questions for the candidate

- Q1. How to interpret Exhibit 1 and what are your insights?
- Note for interviewer: Some OEMs also sell data systems to businesses. Such OEMs are called “Enterprise OEM”. Building own sales force will change the current supplier-buyer relationship with Enterprise OEMs to direct competitor relationship. Keep pressing candidate until she/he gets to this. Then give Exhibit 2.

Solution Guide

- Key insights
 - While OEMs offer excellent channels for our client, they also charge a 25% discount of our list price, leaving us only 7% profit margin.
 - The profit margin via own sales force is 15%, but client probably can’t sell as many equipment as they could through OEMs, because client is building sales force from scratch and lacks relevant experience.
- Most critical insight
 - If client sell via own sales force, they turn themselves into direct competitors against some OEMs!

Exhibit 2: Delson's revenue & profit margin

Total revenue	\$20B
Overall profit margin	6%
Profit Margin on Enterprise OEMs	10%
Enterprise OEMs' share of total revenue	20%
Top 3 Enterprise OEMs' share of total Enterprise OEMs revenue	40%

Note: top 3 Enterprise OEMs will switch to other drive manufacturers if Delson builds own sales force for SSS

Suggested Solution

Questions for the candidate

- Q2. With Exhibit 2, how would you evaluate whether or not to build own sales force?

Solution Guide

- Profit loss due to top 3 Enterprise OEMs switching to other drive manufacturers:
 - $\$20B * 20\% * 40\% * 10\% = \$160M$
- Calculate how many equipment SSS needs to sell to compensate the above loss of hard drive sales profit:
 - Profit margin difference between own sales force and OEM channel is $15\% - 7\% = 8\%$
 - In order to compensate the loss, SSS needs to sell X dollars of equipment via own sales force, then:
 $X * 8\% = 160M \Rightarrow X = \$2B$
- **Insight**: as a new start-up, SSS may not be able to achieve \$2B sales in its first year of sales; therefore, building own sales force is not a good idea.

Suggested Solution

Questions for the candidate

- Q3. If SSS can sell more than \$2B in its first year, how would you change your idea?
- Note for interviewer: can ask candidate to make a final recommendation after this question.

Solution Guide

- Hard drive industry has a very high fixed cost. Major players have very thin profit margin and need to maximize utilization in order to stay profitable.
- Therefore, if client loses top 3 Enterprise OEMs ($\$20B \times 20\% \times 40\% = 1.6B$), their hard drive's profit margin may go down to negative.
 - A rough estimation (just for illustrative purpose here):
 - Current total costs: $\$20B \times (1 - 6\%) = \$18.8B$
 - Revenue after losing top 3 Enterprise OEMs: $\$18.4B$
 - Because client is highly fixed-cost based, the total costs without serving top 3 Enterprise OEMs may be still above 18.4B, leading to negative margin.
- **Insight**: client can't grow the unpredictable SSS start-up at the risk of losing the whole hard drive business.